AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Engineering

Lab Report

Course Title:	COMPUTER NETWORKS		
Course Code:	COE3206	**Section:**	**C**
Semester:	Summer 2025-26	**Degree Program:**	BSc in CSE
Course Teacher:	**DR. RAJARSHI ROY CHOWDHURY**		

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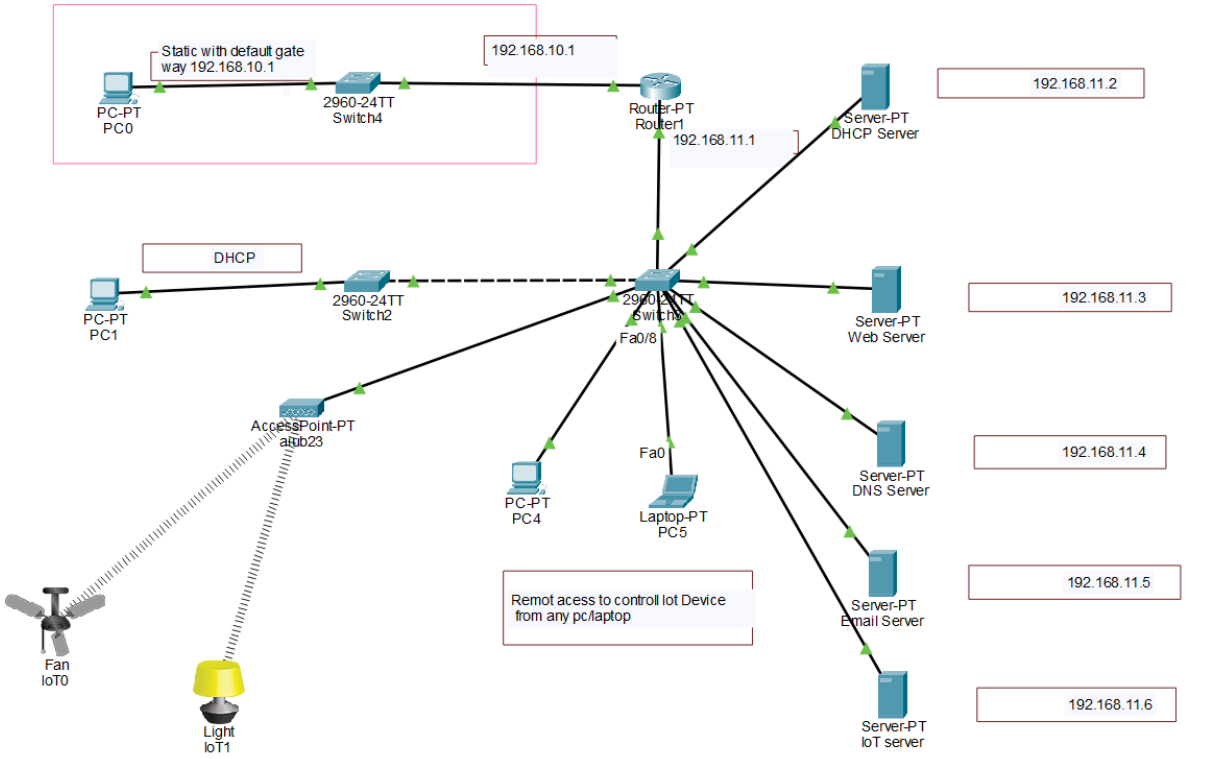
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LAB-1

Configuring a Routed Network with DHCP, DNS, Web, Email Server, and IoT Remote Access Using Cisco Packet Tracer.



1. Introduction

This laboratory exercise demonstrates the design and configuration of a routed computer network using Cisco Packet Tracer. The network shown in the submitted topology contains two IPv4 network segments, a router, multiple switches, client computers, application servers, an Access Point, and IoT devices. The first segment uses 192.168.10.0/24 and the main LAN uses 192.168.11.0/24.

The router provides communication between the two networks. The main LAN contains DHCP, Web, DNS, Email and IoT-related services. Client devices can obtain network parameters automatically through DHCP, resolve a domain name through DNS, access the Web Server, use the Email Server, and remotely monitor/control wireless IoT devices.

4. Network Devices

Device	Role / Function
Router1	Routes traffic between the two IP networks
2960 Switch4	Connects the upper PC network to Router1
2960 Switch2	Connects the PC in the main LAN section
Main 2960 Switch	Central connection point for servers, clients and Access Point
DHCP Server	Provides automatic IP configuration
Web Server	Hosts the test website
DNS Server	Resolves domain names to IP addresses
Email Server	Provides email service
IoT Server	Supports remote IoT monitoring/control
Access Point-PT	Provides wireless connectivity for IoT devices
PC / Laptop	Client and remote-monitoring devices
Fan IoT	Wireless IoT device
Light IoT	Wireless IoT device

5. IP Addressing Scheme

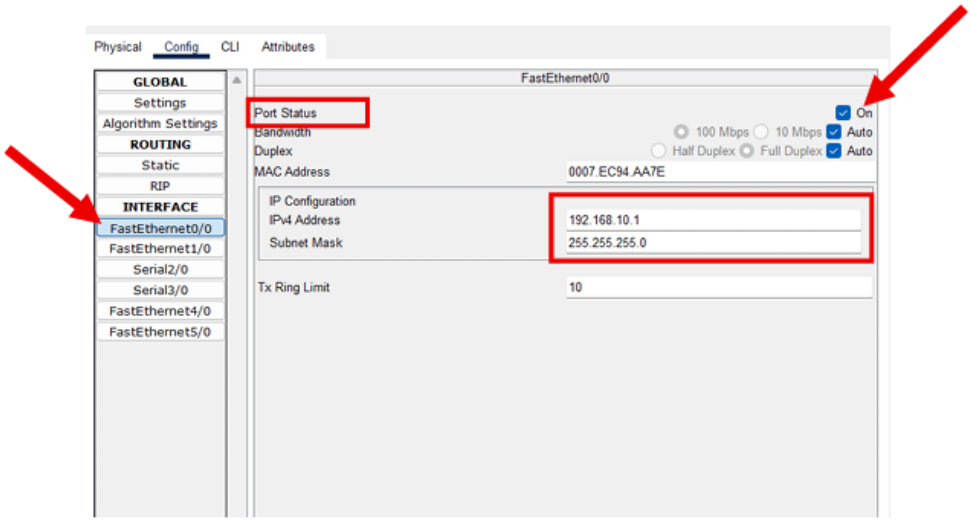
Network / Device	IP Address	Subnet Mask	Gateway
Network 1	192.168.10.0/24	255.255.255.0	192.168.10.1
Router1 – Network 1	192.168.10.1	255.255.255.0	—
Router1 – Main LAN	192.168.11.1	255.255.255.0	—
DHCP Server	192.168.11.2	255.255.255.0	192.168.11.1
Web Server	192.168.11.3	255.255.255.0	192.168.11.1
DNS Server	192.168.11.4	255.255.255.0	192.168.11.1
Email Server	192.168.11.5	255.255.255.0	192.168.11.1
IoT Server	192.168.11.6	255.255.255.0	192.168.11.1
PC1,PC4,PC5	Automatically assigned	255.255.255.0	192.168.11.1
PC0	192.168.10.2	255.255.255.0	192.168.10.1

6. Router Configuration

Router1 is configured as the gateway between the 192.168.10.0/24 and 192.168.11.0/24 networks.

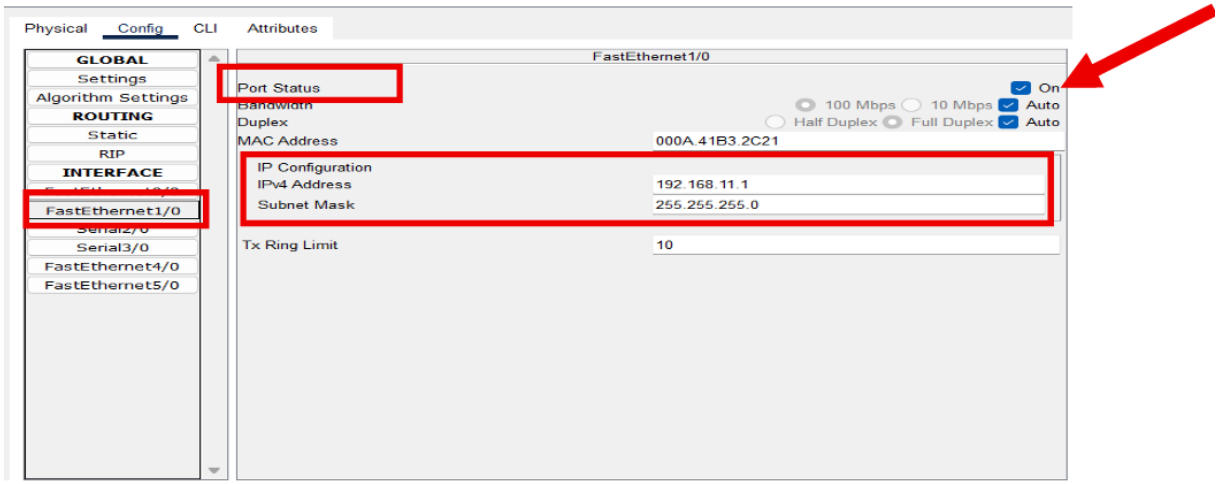
Step-1 First Router Interface

1. Open Router1 and select the Config tab.
2. Select the interface connected to the 192.168.10.0/24 network.
3. Set IP Address to 192.168.10.1.
4. Set Subnet Mask to 255.255.255.0.
5. Turn Port Status On.



Step-2 Second Router Interface

1. Select the interface connected to the main LAN.
2. Set IP Address to 192.168.11.1.
3. Set Subnet Mask to 255.255.255.0.
4. Turn Port Status On.



7. DHCP Server Configuration

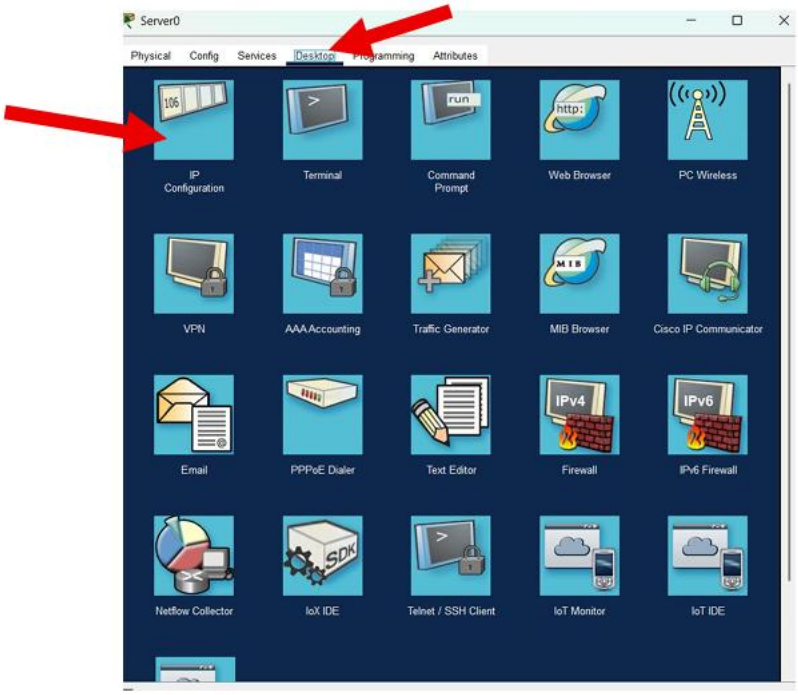
The DHCP Server is configured with the static address 192.168.11.2 so that it can provide network configuration to DHCP clients.

Parameter	**Value**
IP Address	192.168.11.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.11.1
DHCP Service	On
Start IP Address	192.168.11.100

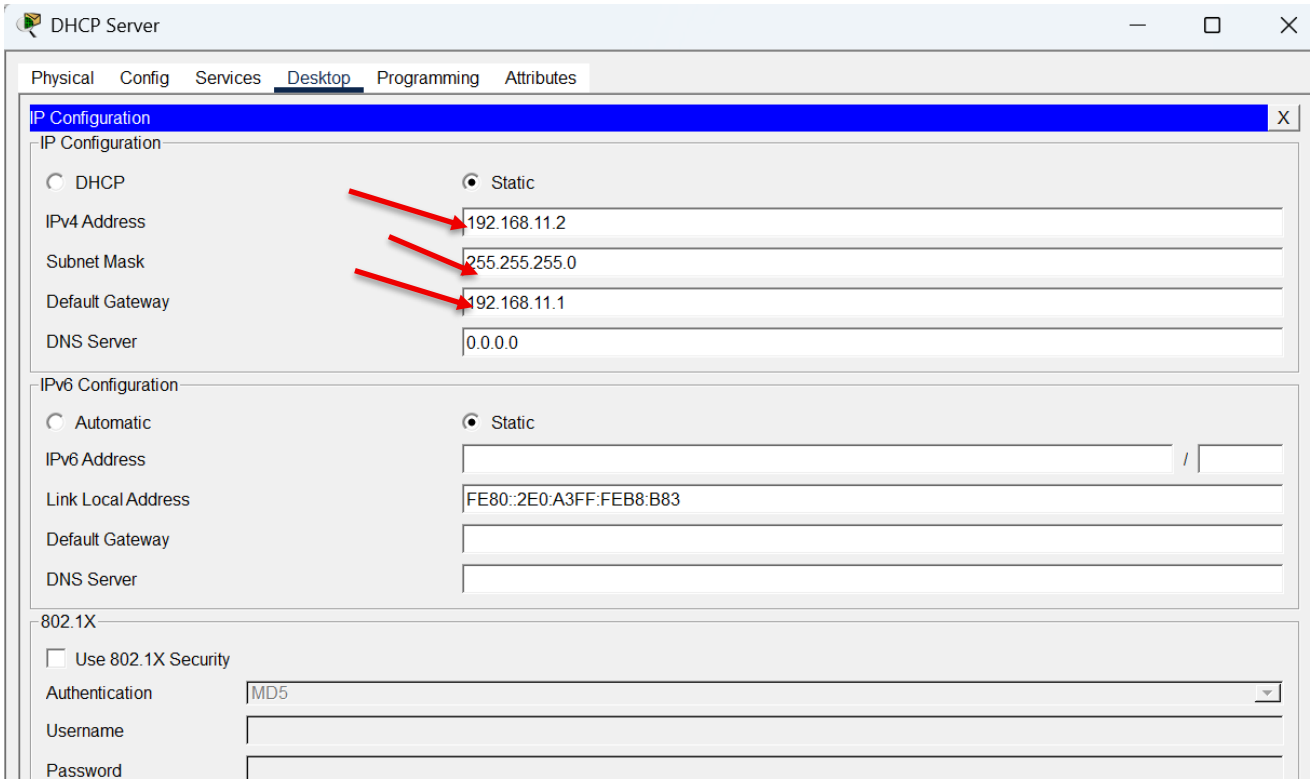
After enabling DHCP, client PCs are configured through Desktop → IP Configuration → DHCP. A successful DHCP request supplies the client with an IP address, subnet mask, gateway and DNS information.

Step 1: Configure Static IP

- Click on Server0.
- Go to "Desktop" tab "IP Configuration".
- As given Picture Below:

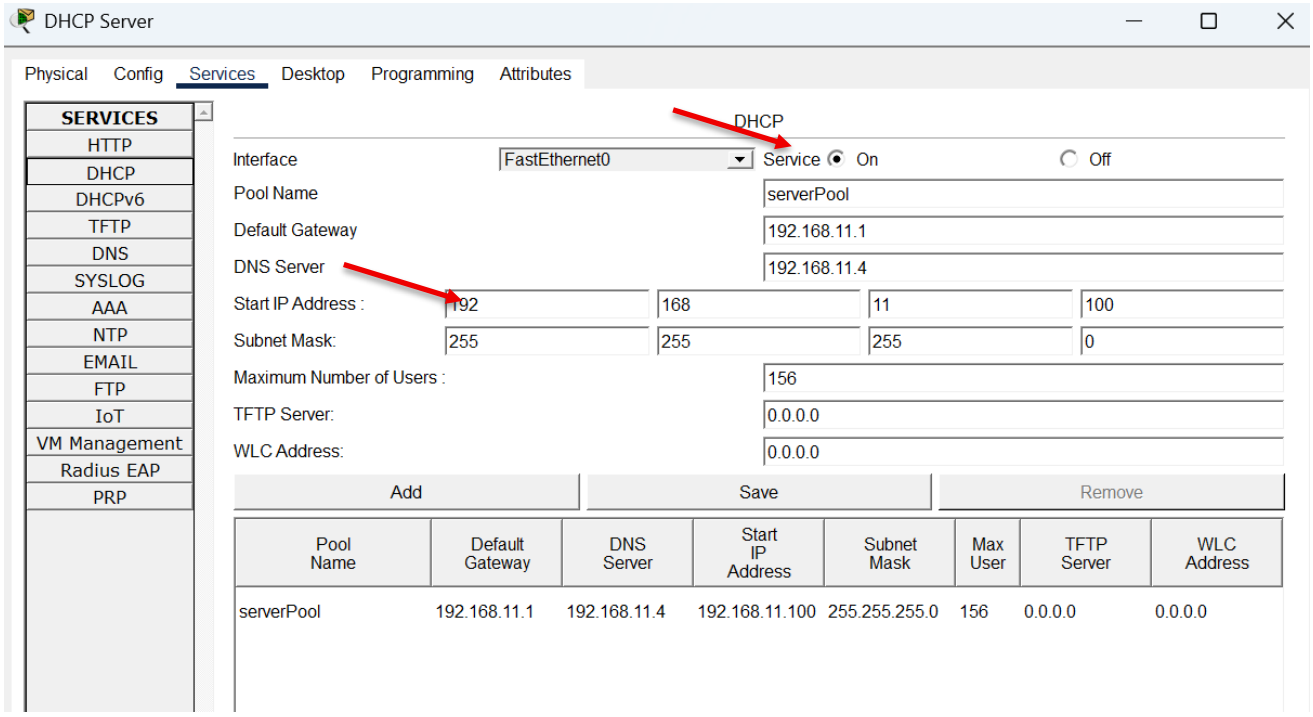


Step -2 :Set IP ,Subnet Mask and Default Gateway



Step-3:

- Set Start Ip :192.168.11.100
- DHCP service ON



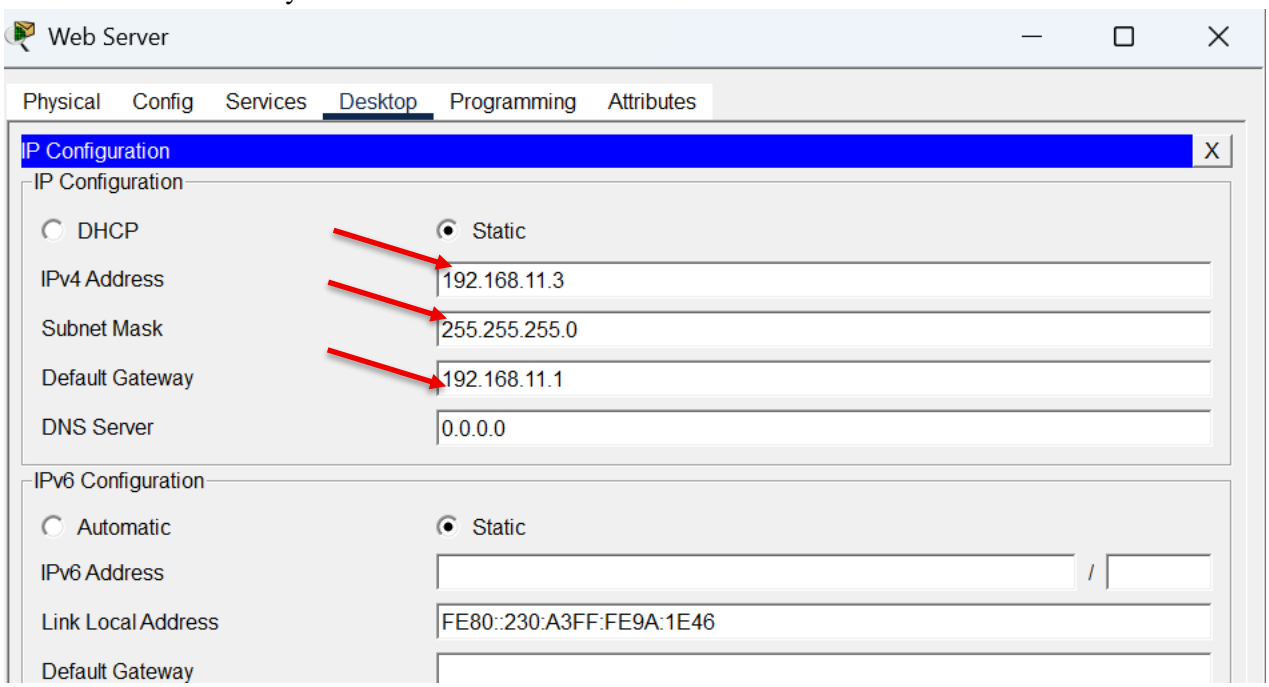
8. Web Server Configuration

The Web Server uses the static address 192.168.11.3/24 with default gateway 192.168.11.1.

8.1 Web Server Setup

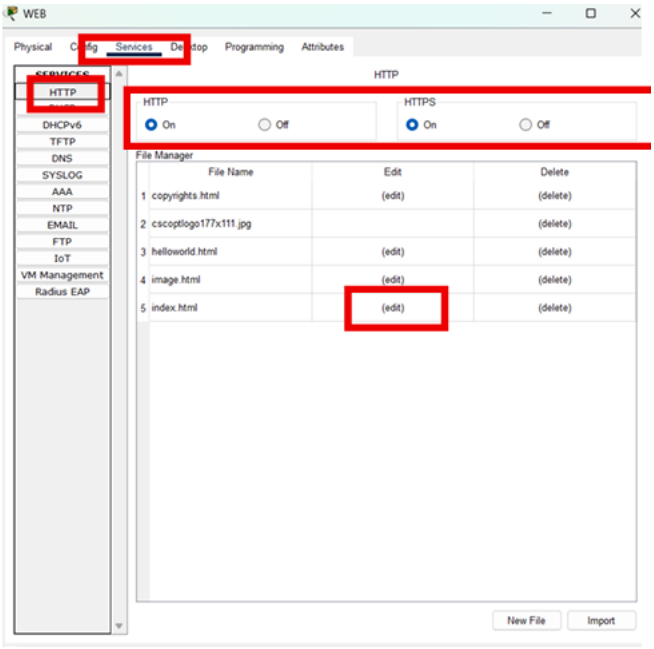
Step-1

- Set Ip -192.168.11.3
- Subnet mask-255.255.255.0
- Default Gateway-192.168.11.1

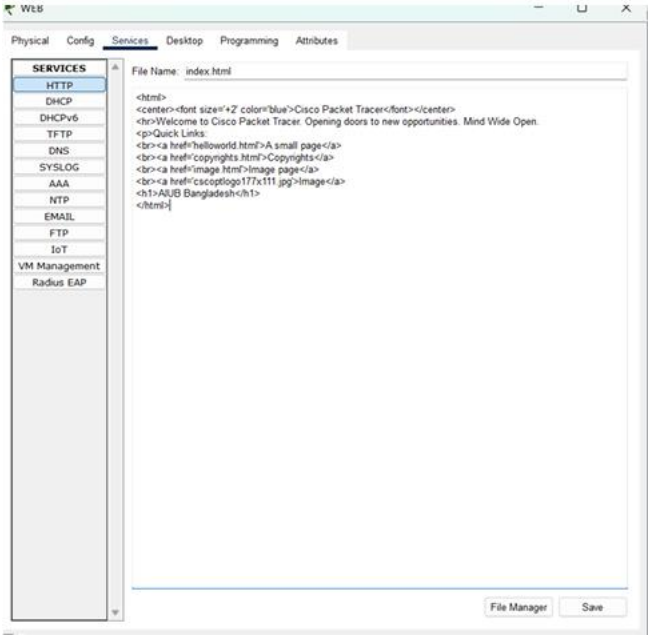


Step 2: Configure WEB Service

- Go to "Services" tab "HTTP".
- ON THE HTTP and HTTPs
- Edit index.html For Test



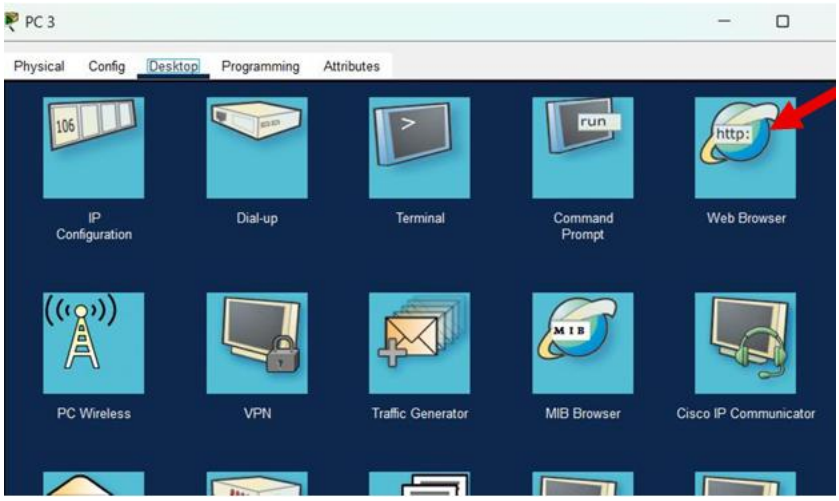
- Give tag <h1>AIUB Bangladesh</h1>



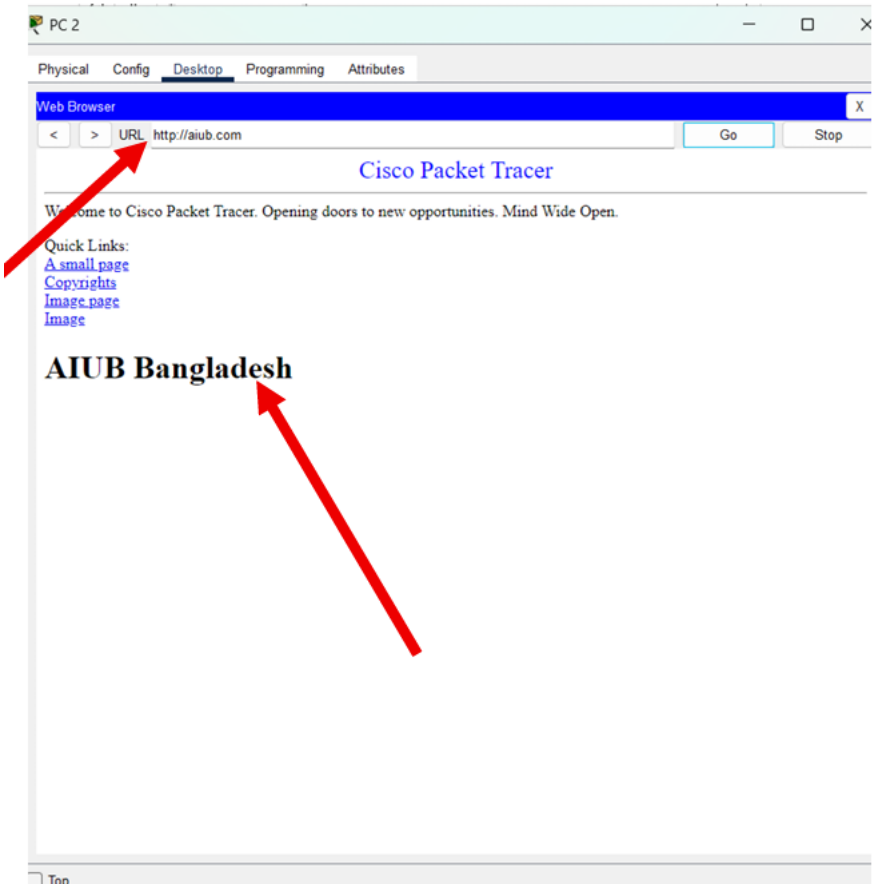
- Then Save
- Click “Add”

8.2 For PC

Desktop-> WebBrowser



Url Search Search: aiub.com &output will be :Aiub Bangladesh



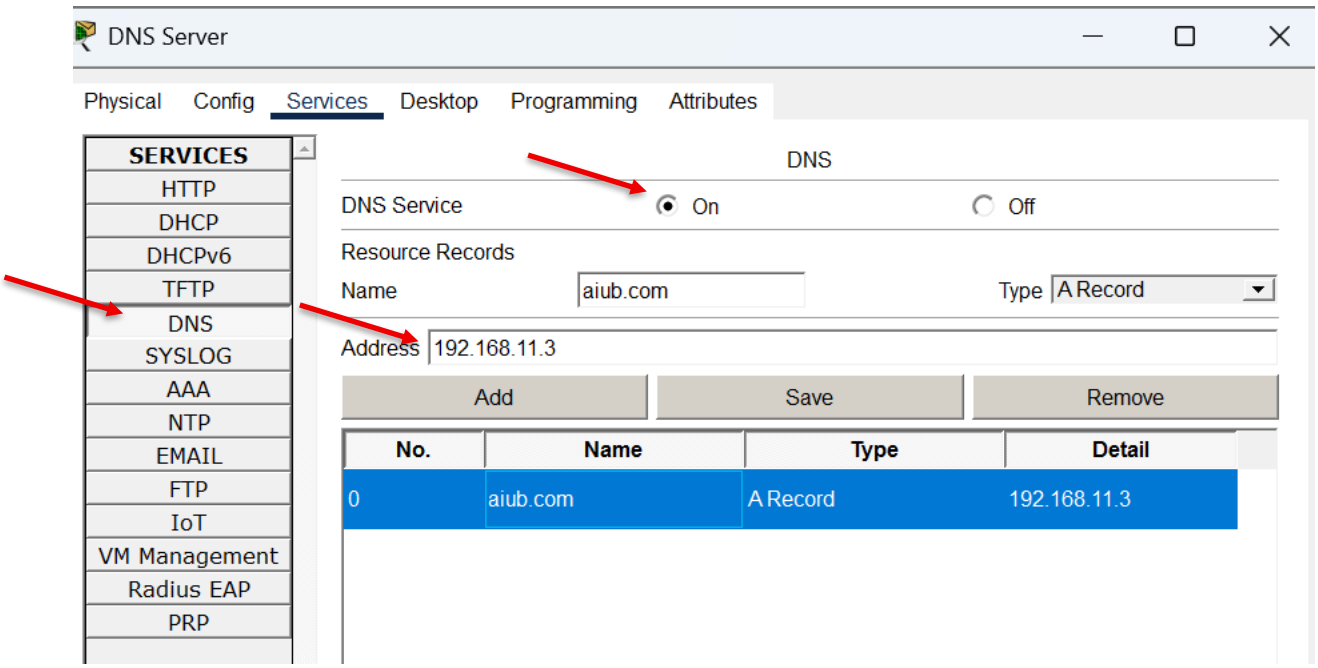
Success!!

9. DNS Server Configuration

The DNS Server uses 192.168.11.4/24 with gateway 192.168.11.1. DNS provides a name-to-IP mapping so that users can access the Web Server using a domain name.

1. Open DNS Server → Services → DNS.

2. Turn the DNS service On.
3. Create the record: aiub.com.
4. Set the corresponding address to 192.168.11.3.
5. Click Add/Save.
6. Configure DHCP clients with DNS Server address 192.168.11.4.

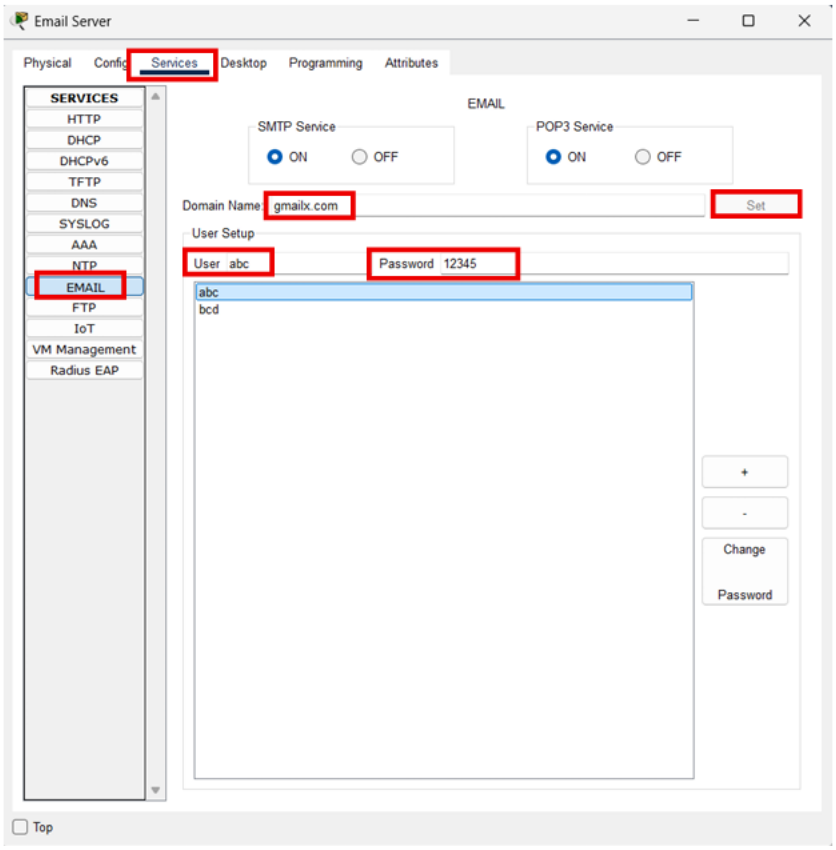


10. Email Server Configuration

The Email Server uses 192.168.11.5/24 with default gateway 192.168.11.1.

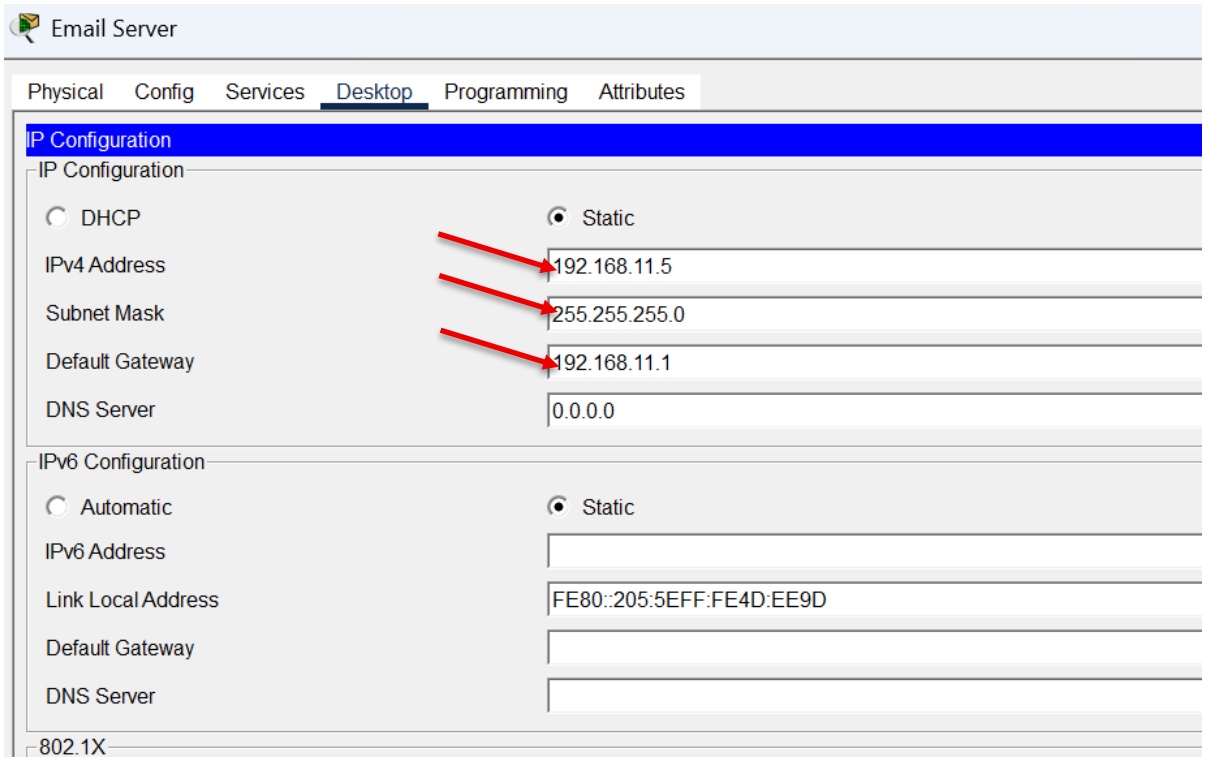
Step-1: Open Email Server Services(as like picture in up)

- Click on the EMAIL
- Give a Domain name as here gmailx.com
- Give user name and password
- Click the set option



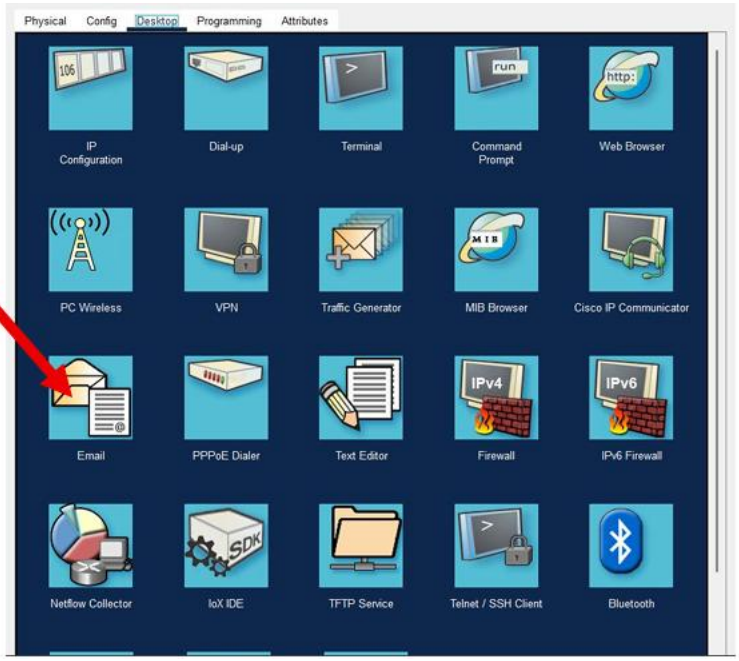
Step 2: Open Email Server Desktop

- Give the IP for this server
- Also get the subnet mask



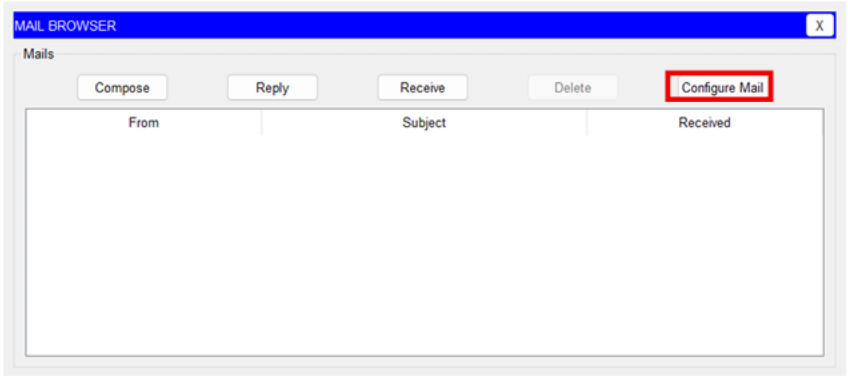
Step 3: Setup PC 4 For email

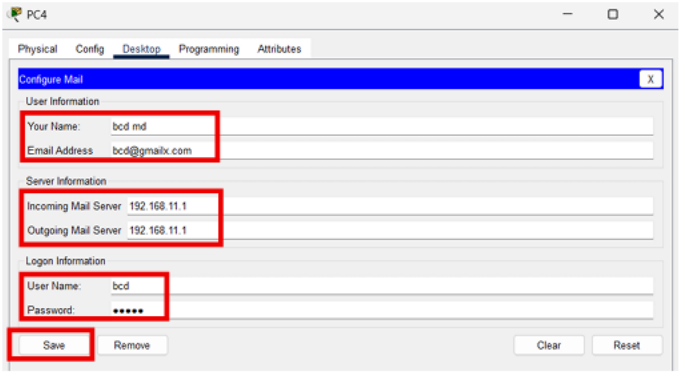
- Click the Email option from desktop
- Click on configure mail o Set any random name
- Enter Email address according to the Email server's username
- For incoming and outgoing Mail server set the same IP as Email server's IP address
- Use same user name and password we used in Email server to create the user o Click save



Step 4: Setup PC 5 For email

- All the steps are same as step 3
- Just use Email server's another user's name & password

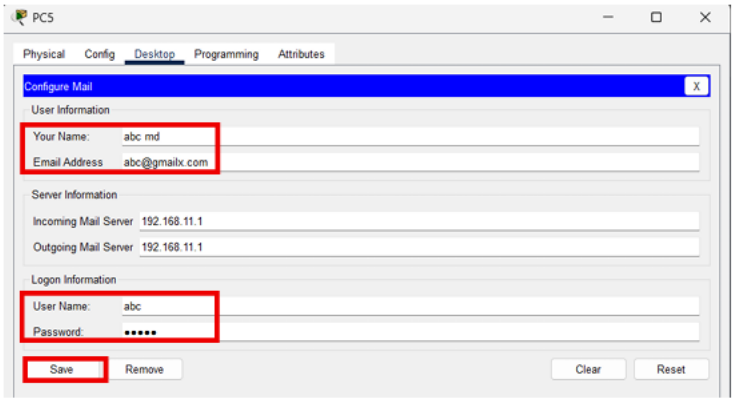




Step 5: Send Email from PC 4

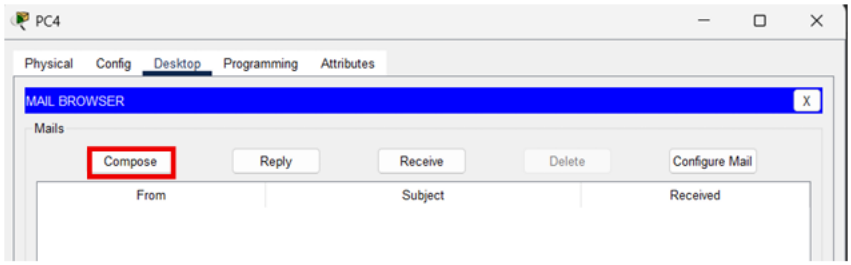
- After clicking on the Email click on compose

- We entered the receiver mail address, Subject & a simple message
- After writing the message we clicked send button
- We will get e success message if all the steps are working perfectly



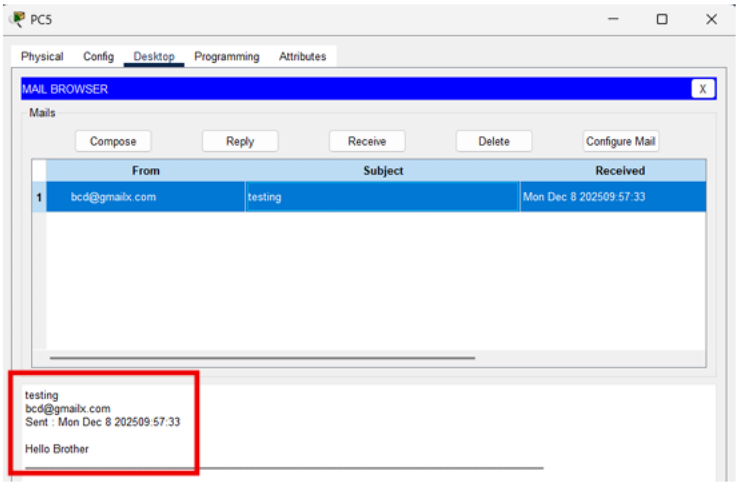
Step 6: Checking Inbox from PC 5

- By clicking on Email we noticed a message that was sent from PC 4
- So, Email server is working perfectly









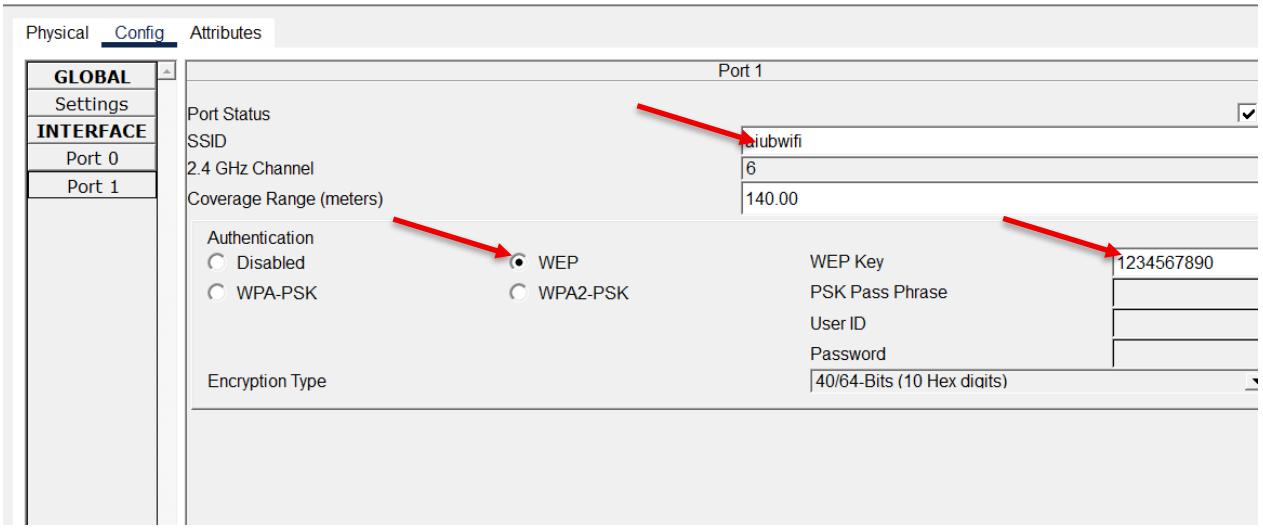
11. Access Point and IoT Configuration

The Access Point provides wireless connectivity for the Fan and Light IoT devices. The Access Point is connected to the main switch using an Ethernet connection.

Wireless Parameter	**Configuration**
SSID	aiubwifi
Authentication	WEP
WEP Key	1234567890

Step-1: Set SSID

- Open Access Point → Config and enable the required Ethernet/wireless interface.
- Set the SSID to aiubwifi

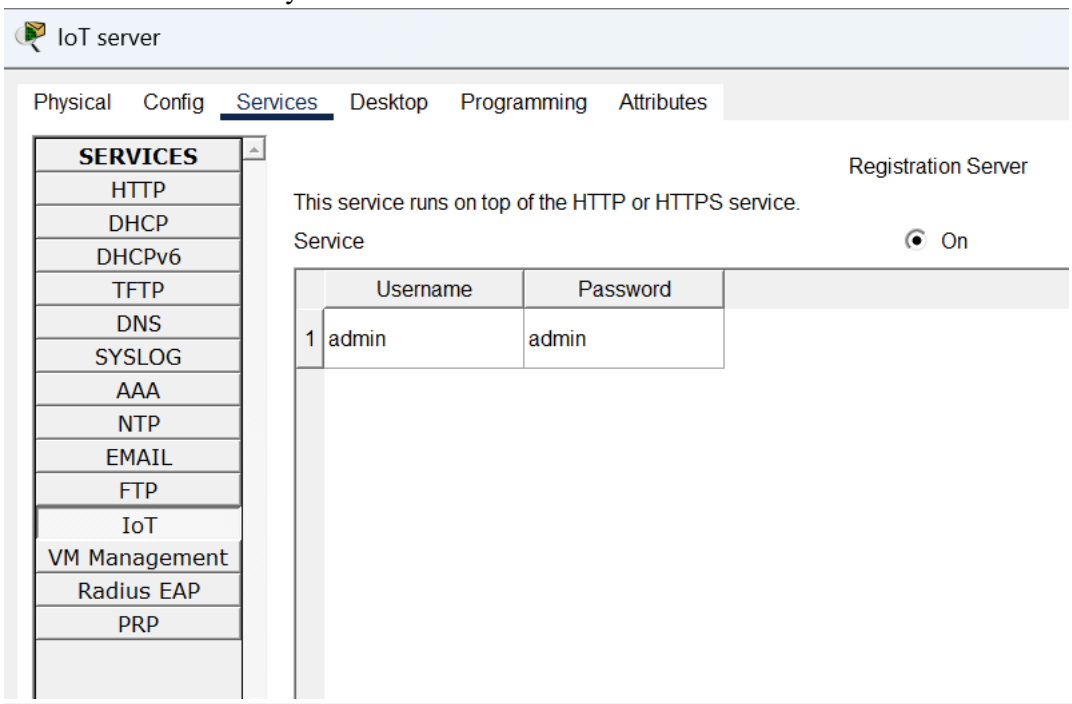


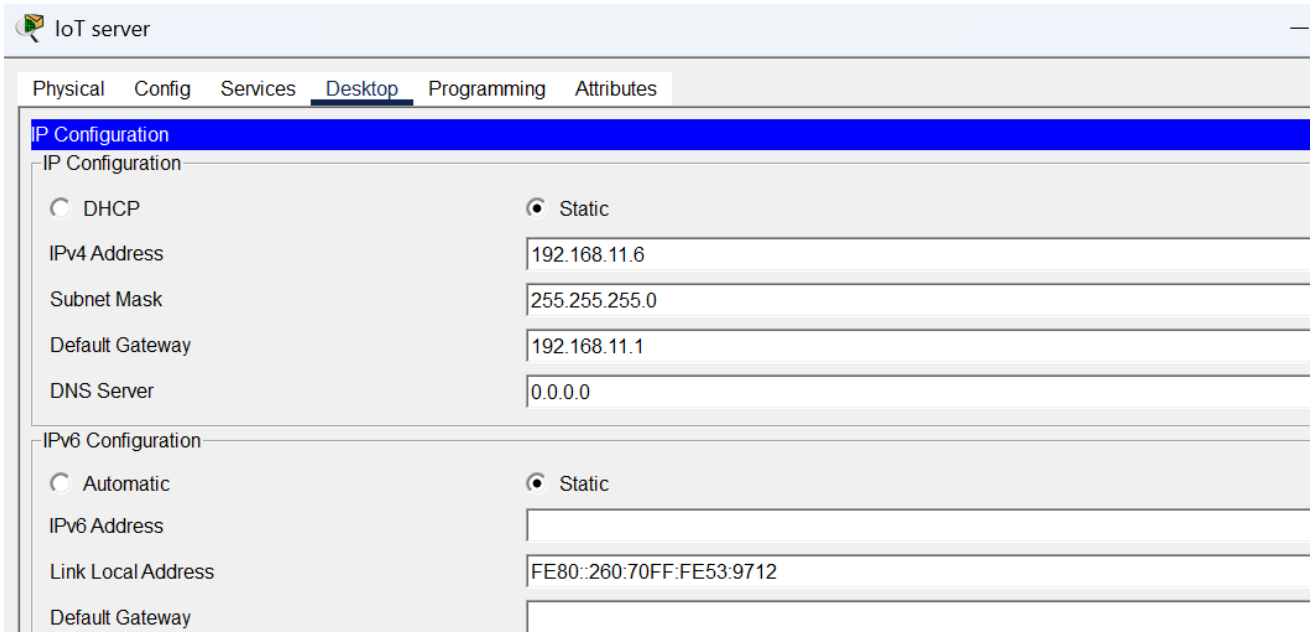
.

- Configure WEP authentication and use the key 1234567890.
- Open each IoT device's wireless configuration.
- Select SSID AIUB-iot.
- Use WEP authentication and the same key.
- Configure the IoT devices to obtain IP configuration through DHCP.

Step -2: Server Sitting

- Set Iot User =admin and Password =admin
- Set ip:192.168.11.6
- Subnet Mask :255.255.255.0
- Default Gateway:192.168.11.1



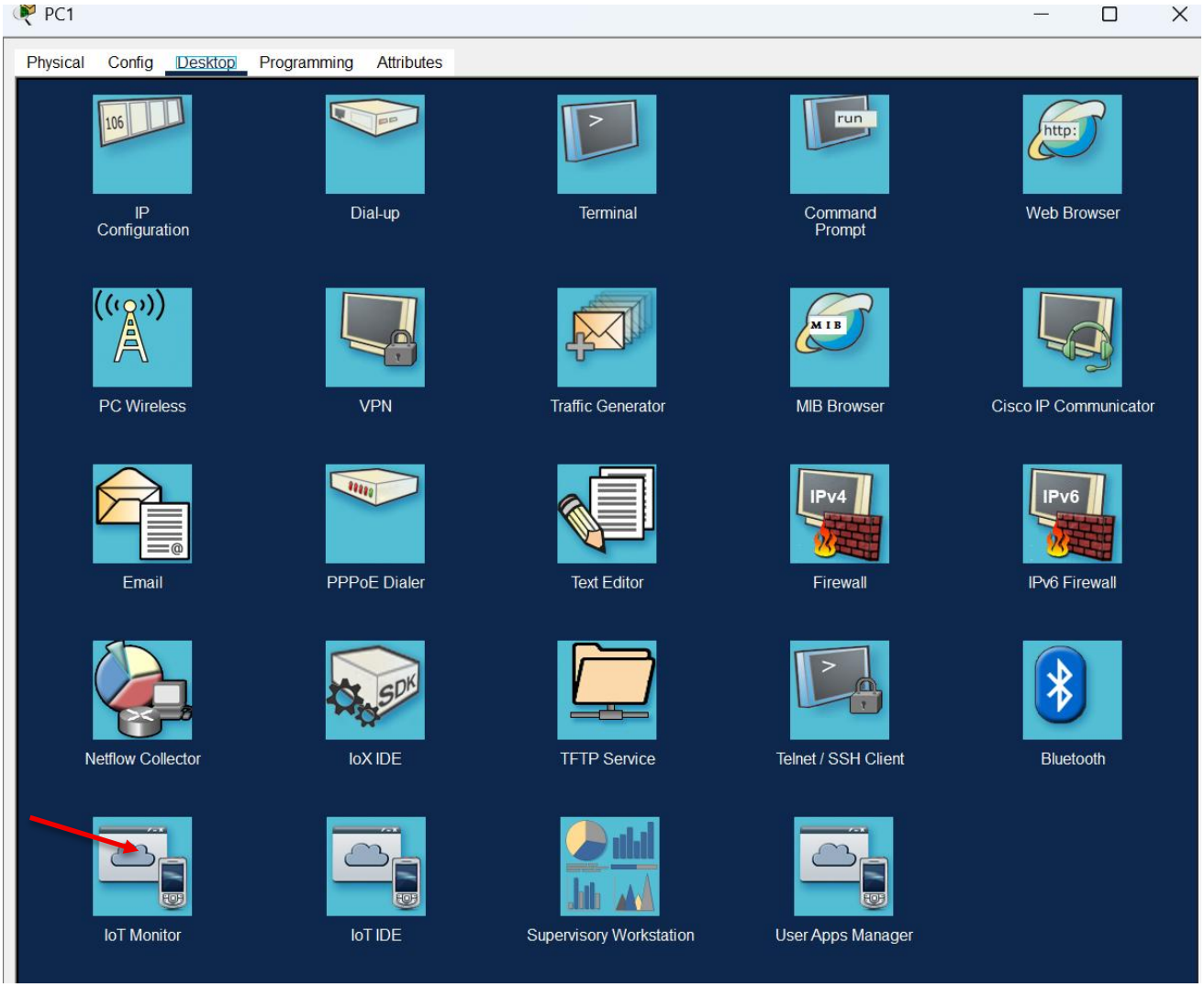


12. Remote IoT Monitoring

The IoT monitoring function allows a PC or laptop to access and control connected IoT devices remotely.

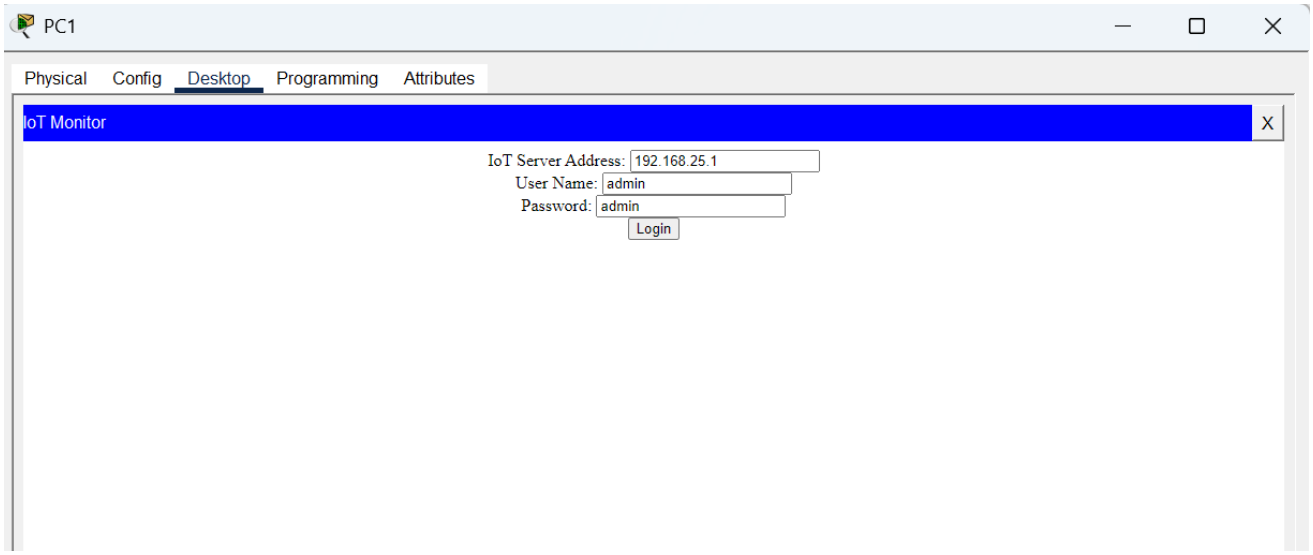
Step-1:

- Open the IoT Monitor application from the client device.



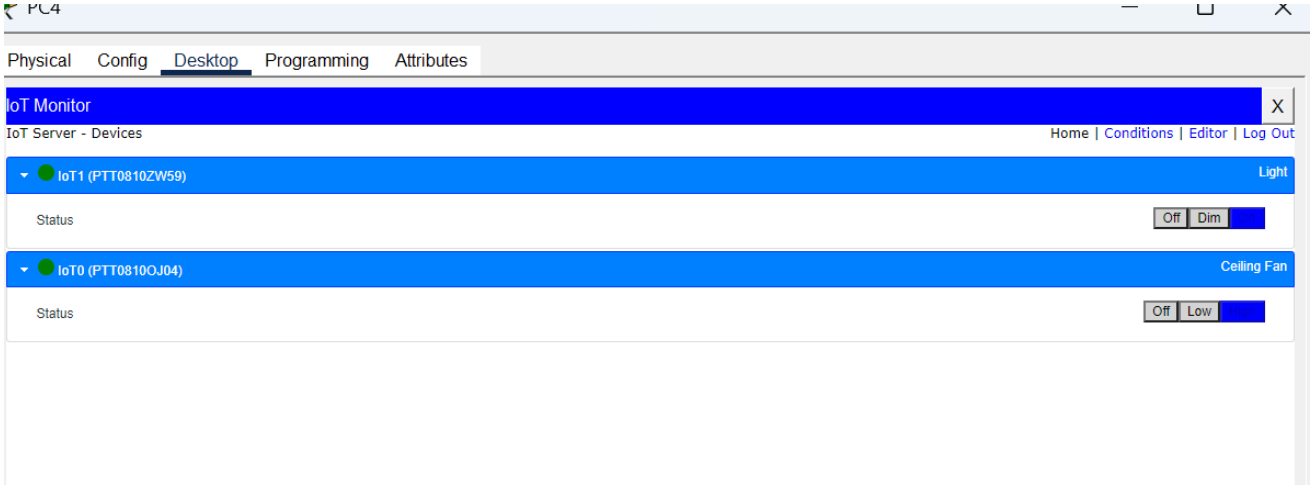
Step-2

- Create/login using the configured IoT monitoring account.
- Connect to the IoT Server using its server address and credentials.



Step:4 –

- Open the device list after successful connection.
- Check the available IoT devices.
- Change the state of the Light and Fan to verify remote control.



LAB-2

VLAN Configuration and Inter-VLAN Routing Using Router-on-a-Stick

In this lab, the network is divided into three separate VLANs so that devices in one department (VLAN) cannot directly see devices in another department, even though they are connected through the same physical switches. Two switches are used, and a router is connected with a single link that is split into sub-interfaces so it can route traffic between all three VLANs (this setup is generally called Router-on-a-Stick).

Network Components

1. 1 Router (Router0)
2. 2 Switches (Switch1 and Switch2)
3. PC2 and PC3 (placed in VLAN 10)
4. PC0, PC1 and Laptop2 (placed in VLAN 20)
5. Laptop0 and Laptop1 (placed in VLAN 30)

Topology Description

Switch2 is the main switch in the middle. Router0 is connected to Switch2 on port Fa0/0. Switch2 is also connected to Switch1 on port Fa0/1, and this link is used to extend VLAN 30 to the second switch. The rest of Switch2's ports are used to connect the VLAN 10 and VLAN 20 devices directly, while Switch1's remaining ports connect the two VLAN 30 laptops.

The IP addressing used for each VLAN is shown below:

VLAN	Network	Router Gateway	Devices in this VLAN
VLAN 10	192.168.10.0/24	192.168.10.1	PC2, PC3
VLAN 20	192.168.20.0/24	192.168.20.1	PC0, PC1, Laptop2
VLAN 30	192.168.30.0/24	192.168.30.1	Laptop0, Laptop1

Step 1: Place the Devices and Connect the Cables

1. Drag Router0, Switch1, Switch2, and all the PCs/Laptops onto the workspace.
2. Connect Router0's Fa0/0 to Switch2's Fa0/0 using a Copper Straight-Through cable.
3. Connect Switch2's Fa0/1 to Switch1's Fa0/1, also with a Copper Straight-Through cable.
4. Connect PC2 and PC3 to Switch2 (any two free ports, e.g., Fa0/3 and Fa0/4).
5. Connect PC0, PC1 and Laptop2 to Switch2 (e.g., Fa0/5, Fa0/6, Fa0/7).
6. Connect Laptop0 and Laptop1 to Switch1 (e.g., Fa0/2 and Fa0/3).

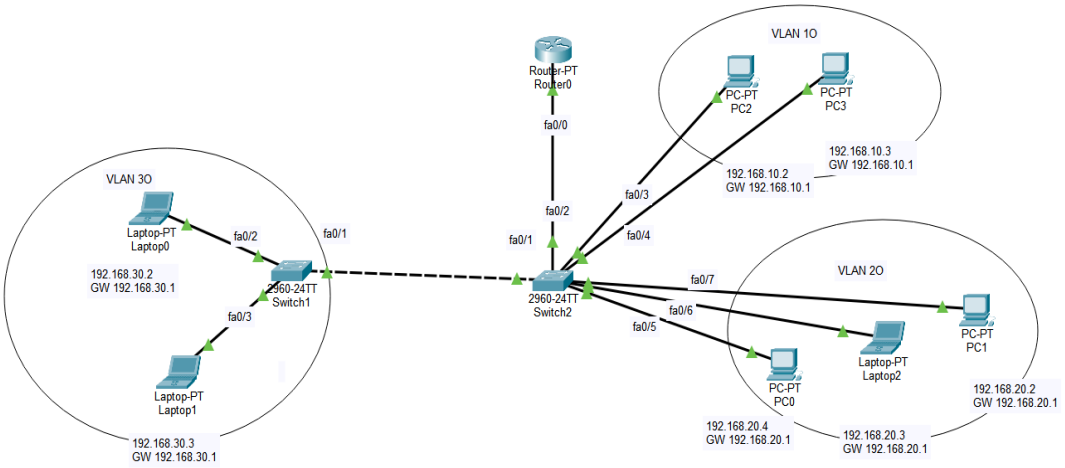
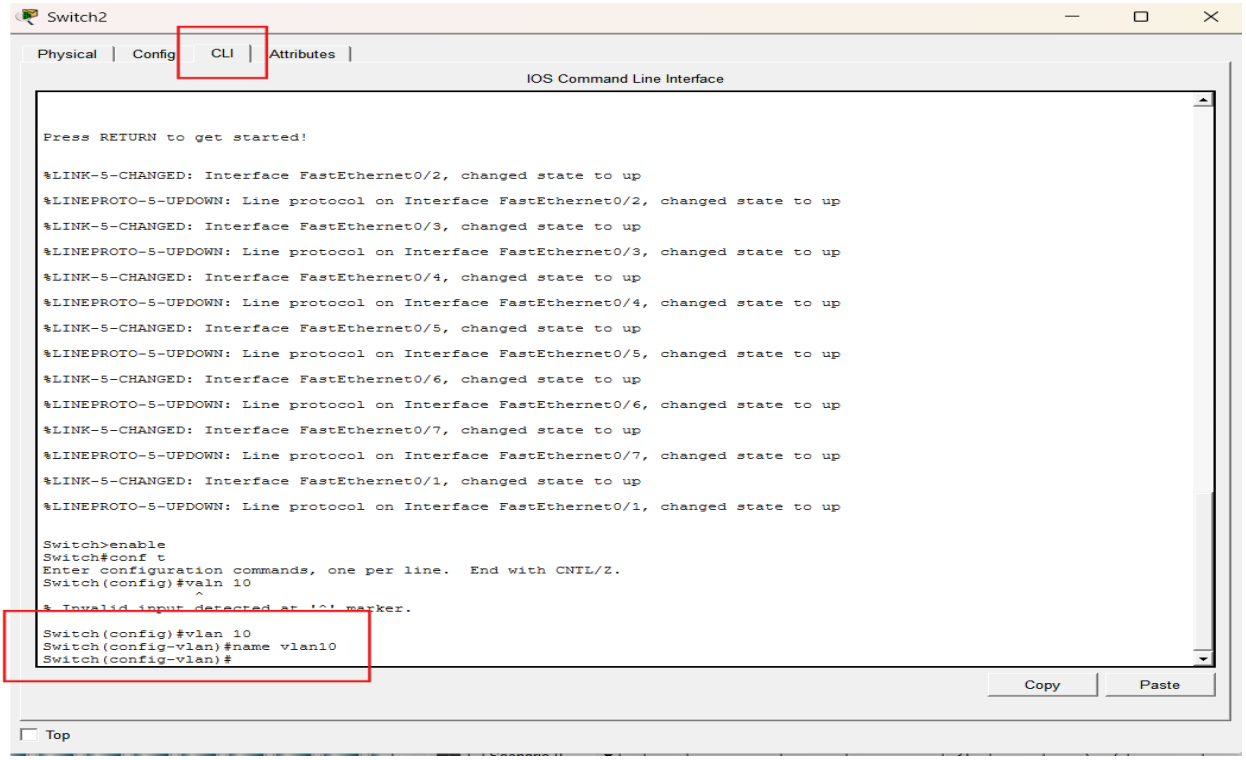


Fig1: Topology diagram

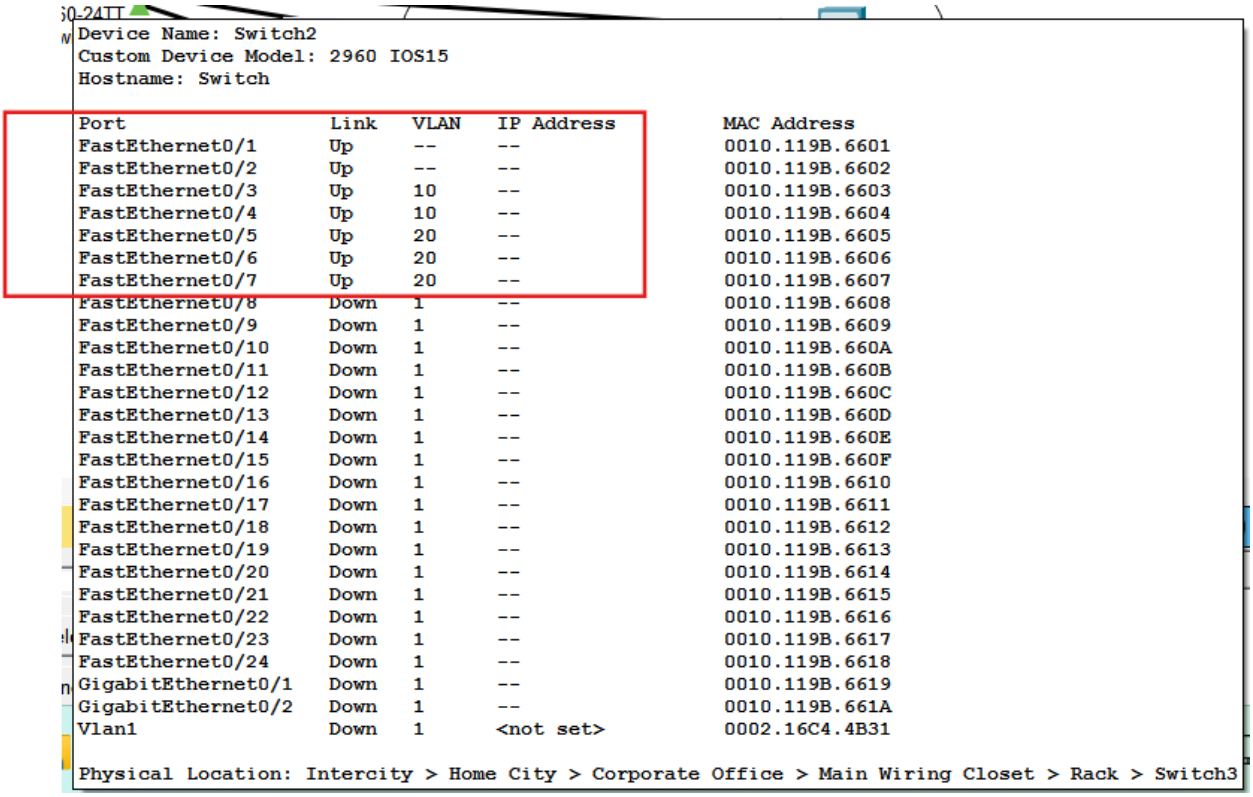
Step 2: Create the VLANs on Switch2 (CLI)

Open Switch2 and click on the CLI tab instead of the Config tab , all the VLAN and port settings in this lab are done directly through the command line rather than the GUI pages.

1. Type enable, then conf t, to enter global configuration mode.
2. Create VLAN 10 (type vlan 10, then name vlan10).



3. Create VLAN 20, the same way (vlan 20, then name vlan20).
4. Create VLAN 30, the same way (vlan 30, then name vlan30).
5. Type exit, then show vlan brief to confirm all three VLANs now appear in the VLAN table.



```
Device Name: Switch2
Custom Device Model: 2960 IOS15
Hostname: Switch

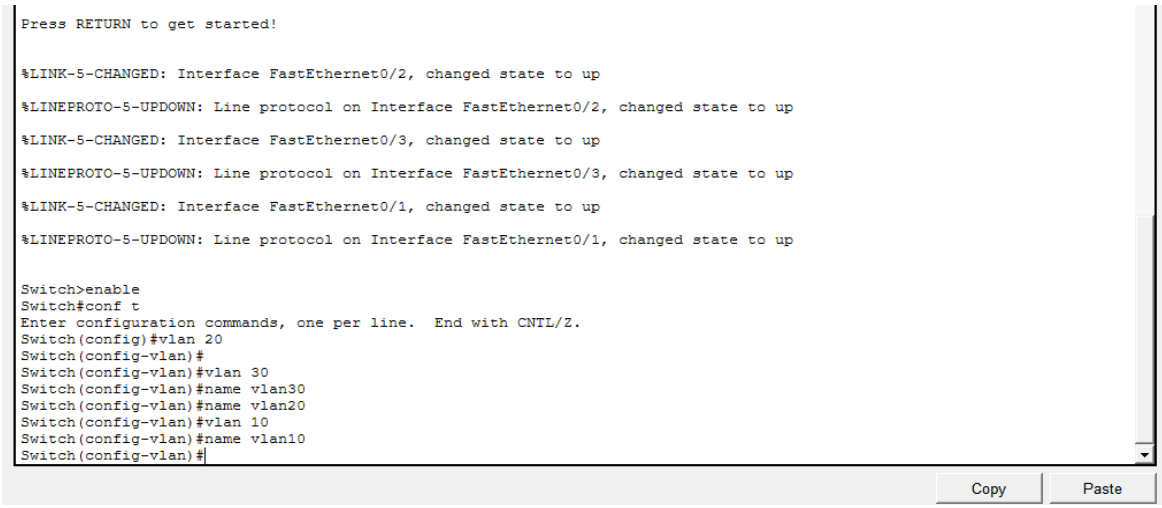
Port                 Link   VLAN   IP Address        MAC Address
FastEthernet0/1      Up     --     --                0010.119B.6601
FastEthernet0/2      Up     --     --                0010.119B.6602
FastEthernet0/3      Up     10     --                0010.119B.6603
FastEthernet0/4      Up     10     --                0010.119B.6604
FastEthernet0/5      Up     20     --                0010.119B.6605
FastEthernet0/6      Up     20     --                0010.119B.6606
FastEthernet0/7      Up     20     --                0010.119B.6607
FastEthernet0/8      Down   1      --                0010.119B.6608
FastEthernet0/9      Down   1      --                0010.119B.6609
FastEthernet0/10     Down   1      --                0010.119B.660A
FastEthernet0/11     Down   1      --                0010.119B.660B
FastEthernet0/12     Down   1      --                0010.119B.660C
FastEthernet0/13     Down   1      --                0010.119B.660D
FastEthernet0/14     Down   1      --                0010.119B.660E
FastEthernet0/15     Down   1      --                0010.119B.660F
FastEthernet0/16     Down   1      --                0010.119B.6610
FastEthernet0/17     Down   1      --                0010.119B.6611
FastEthernet0/18     Down   1      --                0010.119B.6612
FastEthernet0/19     Down   1      --                0010.119B.6613
FastEthernet0/20     Down   1      --                0010.119B.6614
FastEthernet0/21     Down   1      --                0010.119B.6615
FastEthernet0/22     Down   1      --                0010.119B.6616
FastEthernet0/23     Down   1      --                0010.119B.6617
FastEthernet0/24     Down   1      --                0010.119B.6618
GigabitEthernet0/1   Down   1      --                0010.119B.6619
GigabitEthernet0/2   Down   1      --                0010.119B.661A
Vlan1                Down   1      <not set>         0002.16C4.4B31

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch3
```

Step 3: Create the Same VLANs on Switch1 (CLI)

Switch1 also needs these VLANs defined locally, otherwise the trunk link between the two switches will not forward VLAN 30 traffic correctly.

1. Open Switch1's CLI tab and repeat the same commands: enable, conf t, vlan 10, name vlan10, vlan 20, name vlan20, vlan 30, name vlan30.



```
Press RETURN to get started!


%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up


Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 20
Switch(config-vlan)#
Switch(config-vlan)#vlan 30
Switch(config-vlan)#name vlan30
Switch(config-vlan)#name vlan20
Switch(config-vlan)#vlan 10
Switch(config-vlan)#name vlan10
Switch(config-vlan)#
```

Copy Paste

2. Type exit, then show vlan brief to confirm VLAN 10, 20 and 30 are all present on Switch1 as well.

```
Device Name: Switch1
Custom Device Model: 2960 IOS15
Hostname: Switch

Port                Link   VLAN   IP Address        MAC Address
FastEthernet0/1     Up     --     --                0001.97D7.2301
FastEthernet0/2     Up     30     --                0001.97D7.2302
FastEthernet0/3     Up     30     --                0001.97D7.2303
FastEthernet0/4     Down   1      --                0001.97D7.2304
FastEthernet0/5     Down   1      --                0001.97D7.2305
FastEthernet0/6     Down   1      --                0001.97D7.2306
FastEthernet0/7     Down   1      --                0001.97D7.2307
FastEthernet0/8     Down   1      --                0001.97D7.2308
FastEthernet0/9     Down   1      --                0001.97D7.2309
FastEthernet0/10    Down   1      --                0001.97D7.230A
FastEthernet0/11    Down   1      --                0001.97D7.230B
FastEthernet0/12    Down   1      --                0001.97D7.230C
FastEthernet0/13    Down   1      --                0001.97D7.230D
FastEthernet0/14    Down   1      --                0001.97D7.230E
FastEthernet0/15    Down   1      --                0001.97D7.230F
FastEthernet0/16    Down   1      --                0001.97D7.2310
FastEthernet0/17    Down   1      --                0001.97D7.2311
FastEthernet0/18    Down   1      --                0001.97D7.2312
FastEthernet0/19    Down   1      --                0001.97D7.2313
FastEthernet0/20    Down   1      --                0001.97D7.2314
FastEthernet0/21    Down   1      --                0001.97D7.2315
FastEthernet0/22    Down   1      --                0001.97D7.2316
FastEthernet0/23    Down   1      --                0001.97D7.2317
FastEthernet0/24    Down   1      --                0001.97D7.2318
GigabitEthernet0/1  Down   1      --                0001.97D7.2319
GigabitEthernet0/2  Down   1      --                0001.97D7.231A
Vlan1               Down   1      <not set>         00E0.A3A0.21EE

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Switch2
```

Step 4: Set the Trunk Ports (CLI)

A trunk port carries traffic for more than one VLAN at the same time, which is needed on any link between two switches, or between a switch and the router, since those links have to carry all three VLANs together.

1. On Switch2, go to interface Fa0/2 (the port toward Router0): type interface fa0/2, then switchport mode trunk.

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#valn 10
              ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name vlan10
Switch(config-vlan)#int fa0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

Copy Paste

2. On Switch2, do the same for interface Fa0/1 (the port toward Switch1): interface fa0/1, then switchport mode trunk.
3. On Switch1, go to interface Fa0/1 (the port toward Switch2) and set it to trunk as well: interface fa0/1, then switchport mode trunk.

Step 5: Assign the Access Ports (CLI)

Every port that connects directly to a PC or laptop should stay in access mode and be placed in only one VLAN, using switchport mode access and switchport access vlan.

1. On Switch2, configure Fa0/3 and Fa0/4 for VLAN 10: interface fa0/3, switchport mode access, switchport access vlan 10 , then repeat for interface fa0/4.

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#valn 10
               ^
% Invalid input detected at '^' marker.

Switch(config)#vlan 10
Switch(config-vlan)#name vlan10
Switch(config-vlan)#int fa0/2
Switch(config-if)#switchport mode trunk
Switch(config-if)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#
```

Copy Paste

2. On Switch2, configure Fa0/5, Fa0/6 and Fa0/7 for VLAN 20 the same way: interface fa0/5 (then fa0/6, then fa0/7) , switchport mode access, switchport access vlan 20.
3. On Switch1, configure Fa0/2 and Fa0/3 for VLAN 30: interface fa0/2, switchport mode access, switchport access vlan 30 , then repeat for interface fa0/3.

Step 6: Configure Router0 with Sub-interfaces (CLI)

Since Router0 only has one cable going to the switch, it cannot have three separate physical interfaces for the three VLANs. Instead, the single interface Fa0/0 is split into logical sub-interfaces, one per VLAN, using 802.1Q encapsulation. Open Router0's CLI tab and type the following.

1. Type enable, then conf t, to enter global configuration mode.
2. Create the VLAN 10 sub-interface: int fa0/0.10, then encapsulation dot1Q 10, then ip address 192.168.10.1 255.255.255.0

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0.10
Router(config-subif)#encapsulation dot1
% Incomplete command.
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
```

3. Create the VLAN 20 sub-interface: int fa0/0.20, then encapsulation dot1Q 20, then ip address 192.168.20.1 255.255.255.0
4. Create the VLAN 30 sub-interface: int fa0/0.30, then encapsulation dot1Q 30, then ip address 192.168.30.1 255.255.255.0.
5. Finally, bring up the physical interface itself (not a sub-interface): int fa0/0, then no shutdown. The sub-interfaces stay down until this is done.

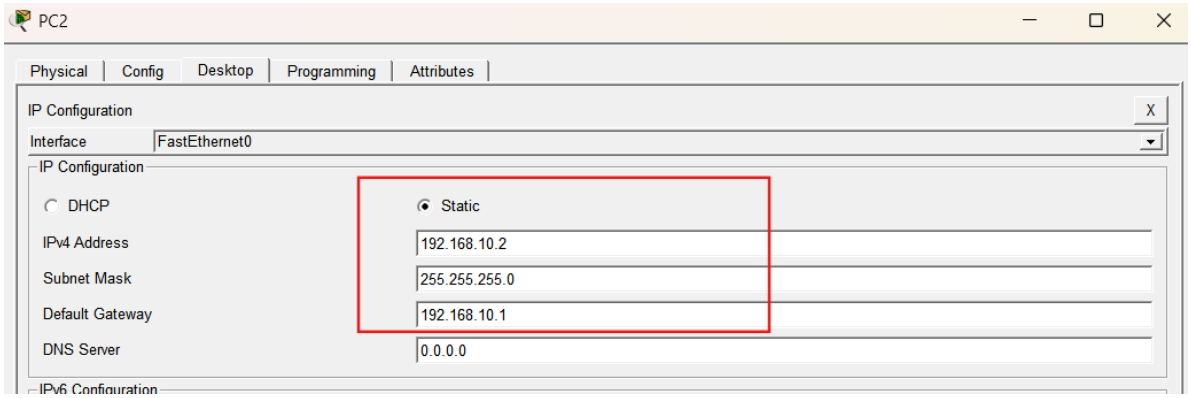
```
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no shutdown
Router(config-if)#
```

Note: when typing the IP address on the router, double-check it carefully (192.168.x.1, not 19.168.x.1) , a single missing digit is a common typing mistake and it will stop the whole VLAN from routing.

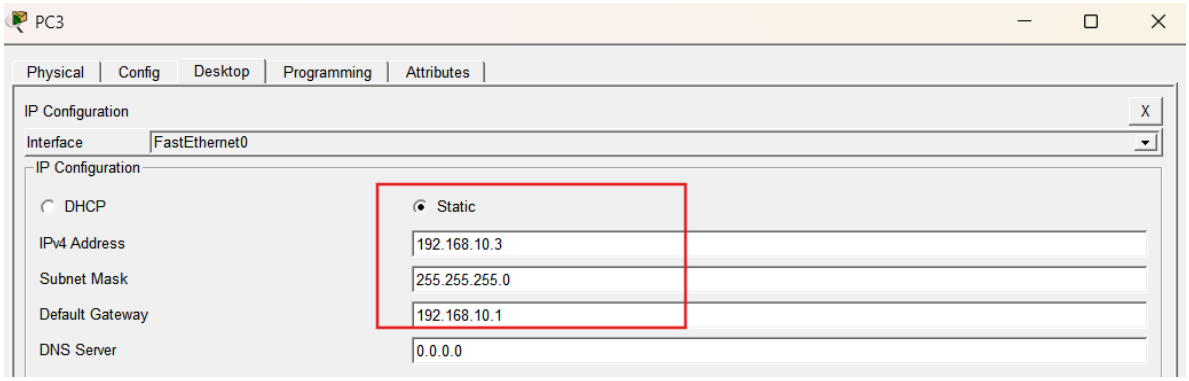
Step 7: Give Each PC and Laptop an IP Address

Each device gets a static IP address from its own VLAN's network, with the matching router sub-interface as the default gateway.

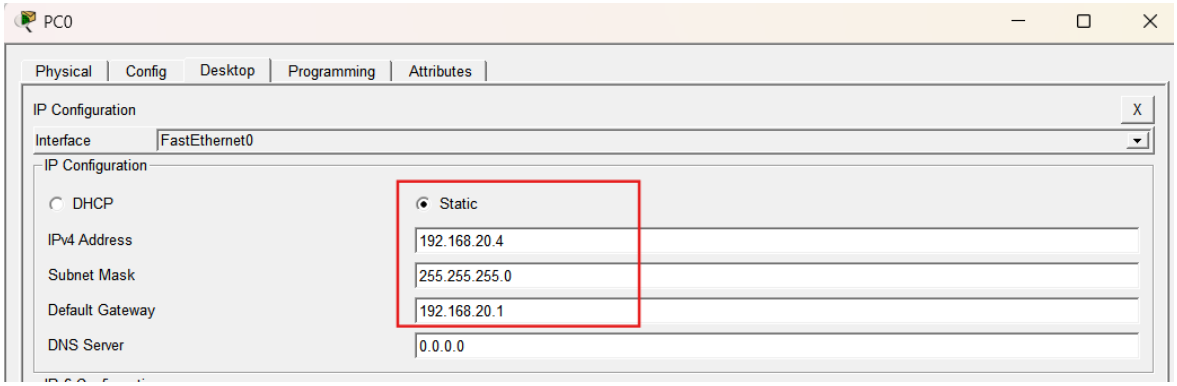
1. PC2: 192.168.10.2 / 255.255.255.0, Gateway 192.168.10.1



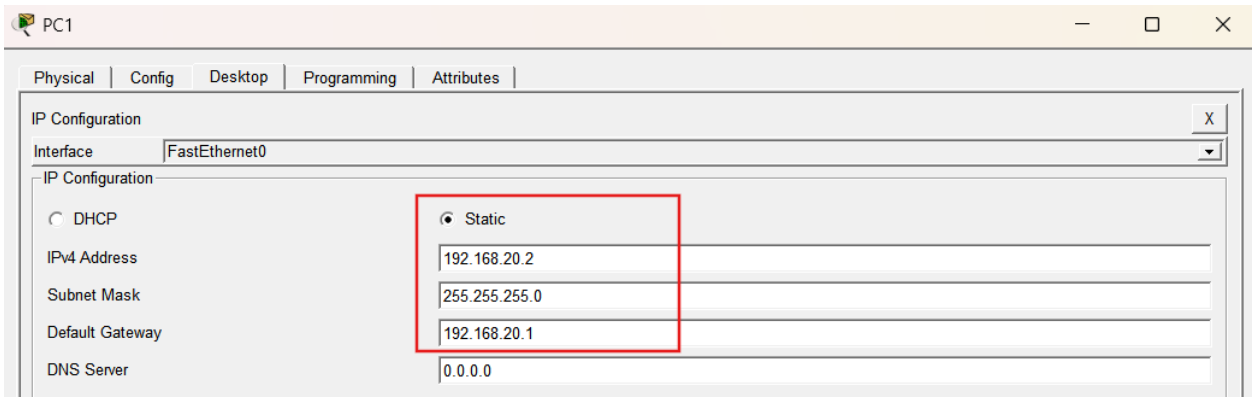
2. PC3: 192.168.10.3 / 255.255.255.0, Gateway 192.168.10.1



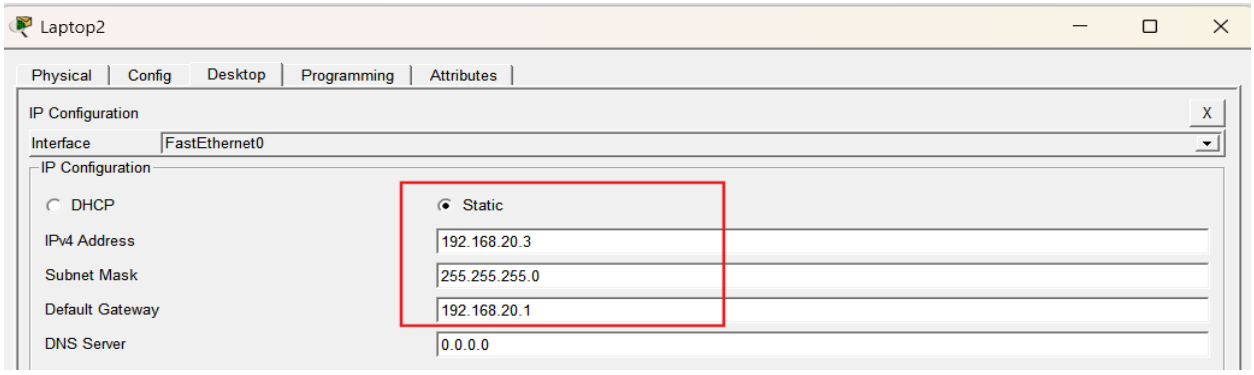
3. PC0: 192.168.20.4 / 255.255.255.0, Gateway 192.168.20.1



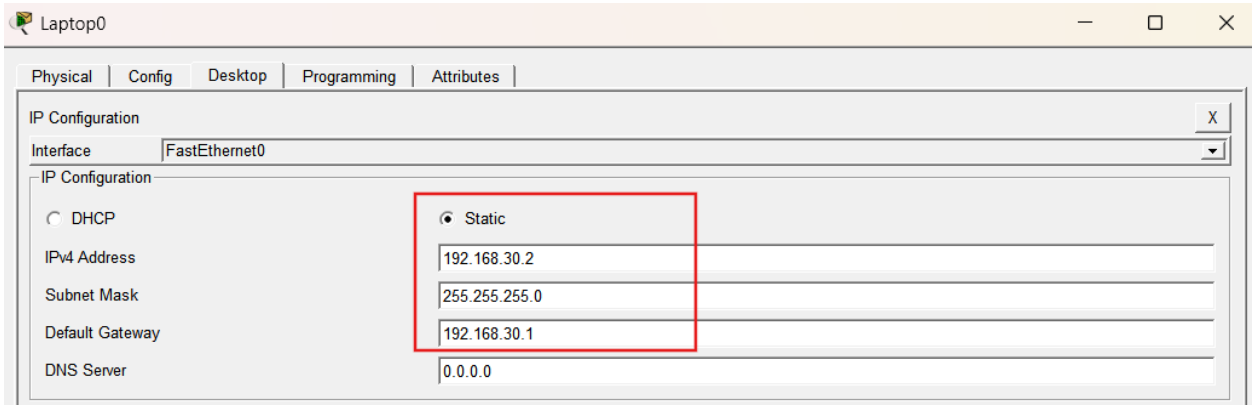
4. PC1: 192.168.20.2 / 255.255.255.0, Gateway 192.168.20.1



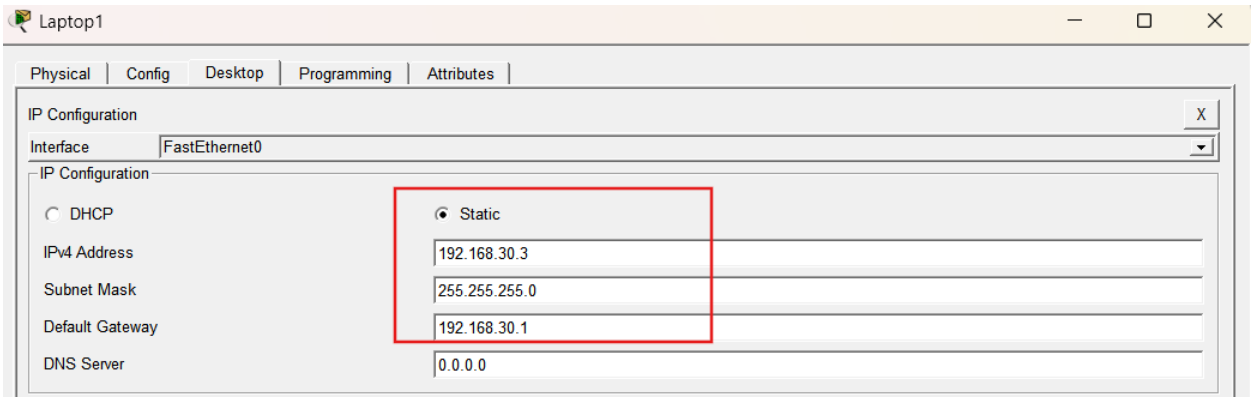
5. Laptop2: 192.168.20.3 / 255.255.255.0, Gateway 192.168.20.1



6. Laptop0: 192.168.30.2 / 255.255.255.0, Gateway 192.168.30.1



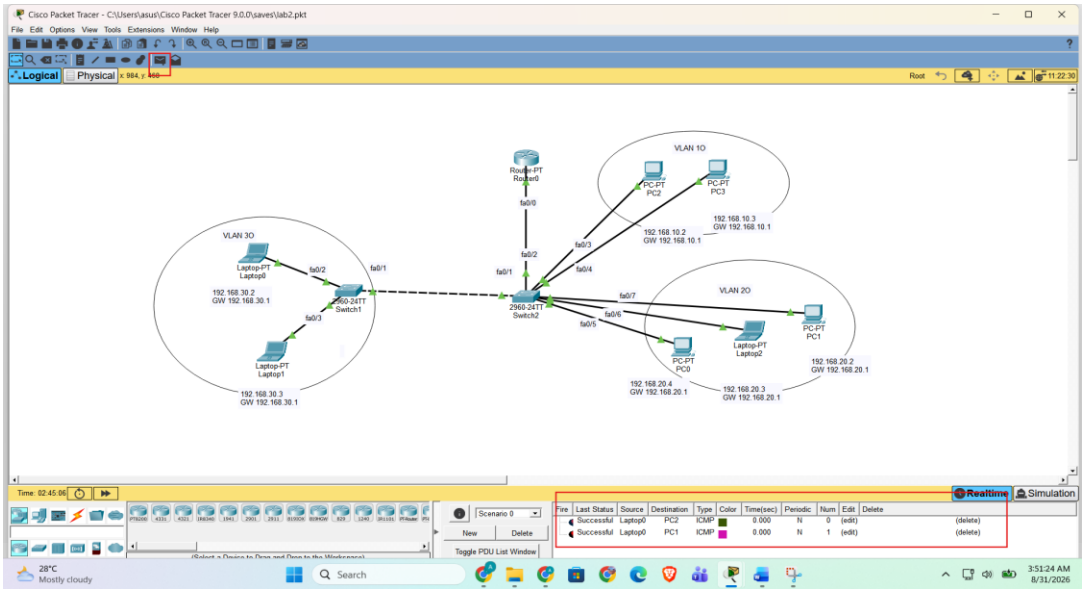
7. Laptop1: 192.168.30.3 / 255.255.255.0, Gateway 192.168.30.1



Step 8: Test the Connection

The last step is to check whether the VLANs and the router sub-interfaces are actually working together correctly. This is done by using the message icon to transfer test packets between devices in different VLANs, while staying in Realtime mode.

1. Make sure the workspace is in Realtime mode (top-right corner), then click the message icon (Add Simple PDU) from the right-side toolbar.
2. Click on a device from one VLAN, for example PC0, and then click on a device from a different VLAN, for example PC2 or PC3 , this transfers a test packet from the first device to the second.
3. Repeat this for a few other device pairs across different VLANs to test more paths (for example PC0 to Laptop0, or PC1 to PC3).
4. Look at the Event List at the bottom of the window and check the Last Status column for each packet ,it should say Successful if the VLANs, trunk links and router sub-interfaces are all set up correctly. If any packet fails, re-check the trunk ports and the sub-interface IP addresses first, since those are the most common mistakes.



Once these packets are all shown as Successful, the VLAN setup and inter-VLAN routing are working as intended, and the lab can be saved.

LAB-3

Introduction

Computer networks allow devices to communicate and share information within and between different networks. Switches connect devices within a local area network, while routers forward packets between separate networks. This experiment uses Cisco Packet Tracer to design and configure a network consisting of three Cisco 2911 routers, two switches, two PCs, and two laptops.The network connects two LANs, 192.168.10.0/24 and 192.168.13.0/24, through the intermediate networks 192.168.11.0/24 and 192.168.12.0/24. The configuration includes selecting suitable cables, assigning static IPv4 addresses, setting subnet masks and default gateways, and enabling router interfaces. Routing Information Protocol version 2 (RIPv2) is configured through the Command-Line Interface (CLI) to allow the routers to exchange routing information and learn routes to remote networks automatically.The main objective of this experiment is to establish end-to-end communication between devices on different LANs. Connectivity is checked using ping commands and Simple PDU tests. This lab develops practical understanding of IP addressing, router configuration, dynamic routing, and basic network troubleshooting.

Required Devices: -

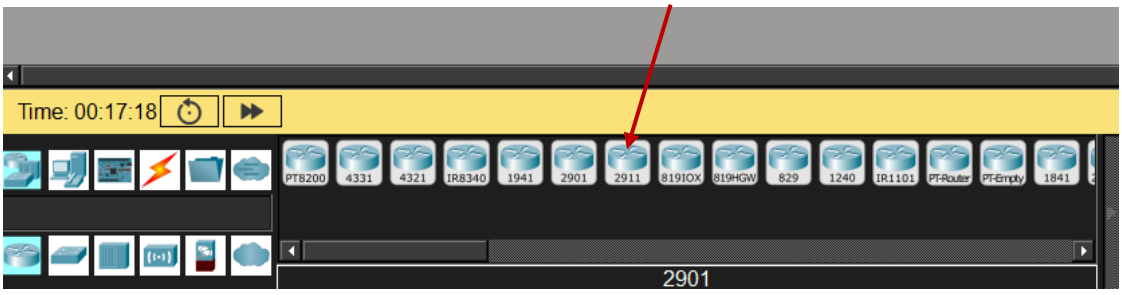
- 2 Laptops
- 2 Pcs
- 3 (2911Router)
- 2 (2760-24TT Switch)
- 2 Copper Cross-Over Cable
- 6 Copper Cable

1. **Spep-1:**



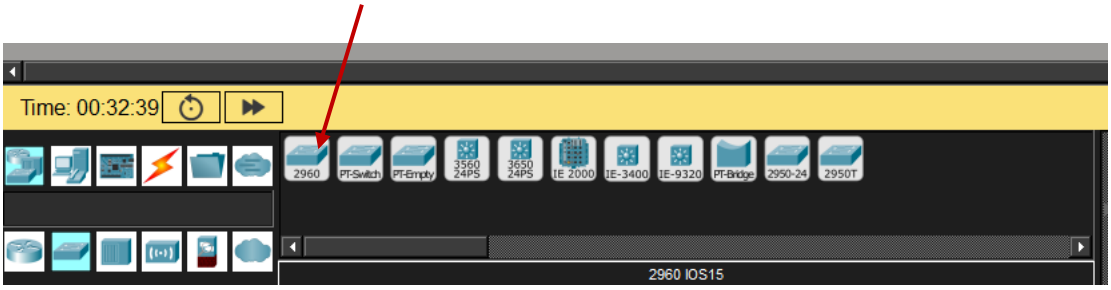
Add 2 Pc and 2 Laptop from this part.

2. **Step-2:**



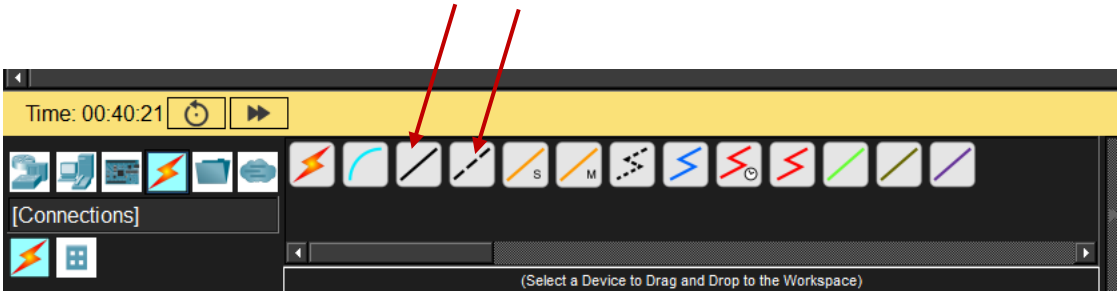
Add 3 (2911Router) from this part

3. **Step-3:**



Add 2 (2760-24TT Switch) from this part

4. **Step-4:**



From this part we select cable to connect the component. In this lab we use 2 types of Cable. One is Copper Cross-Over Cable's and another is Normal Copper Cable's.

For Router to Router, we use **Copper Cross-Over Cable's** Like this: -

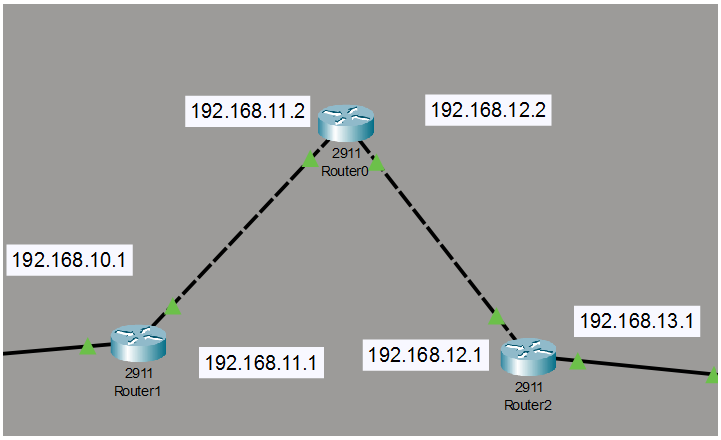


Fig-1: Router to router connection.

For Pc/Laptop to Switch, and Switch to Router we use **Normal Copper Cable's.** Like this: -

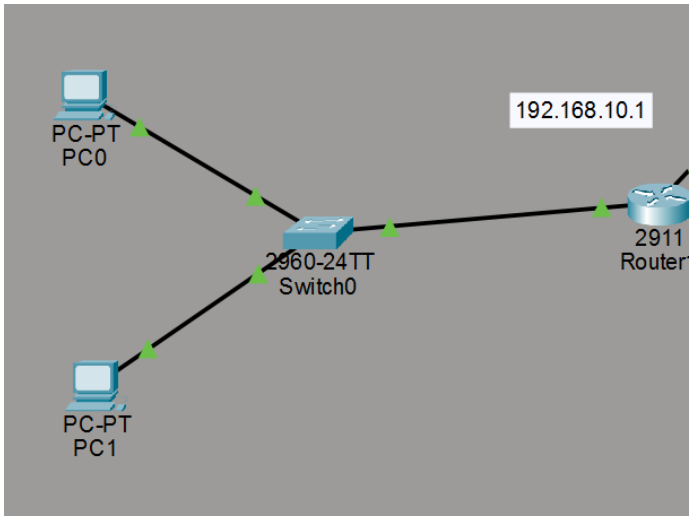
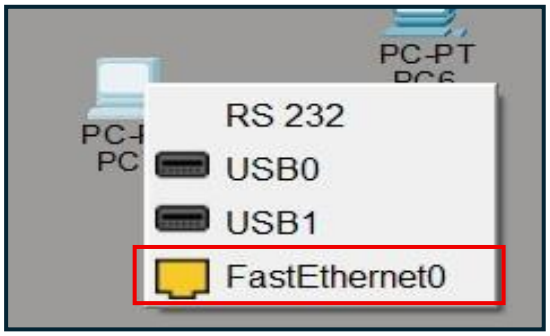
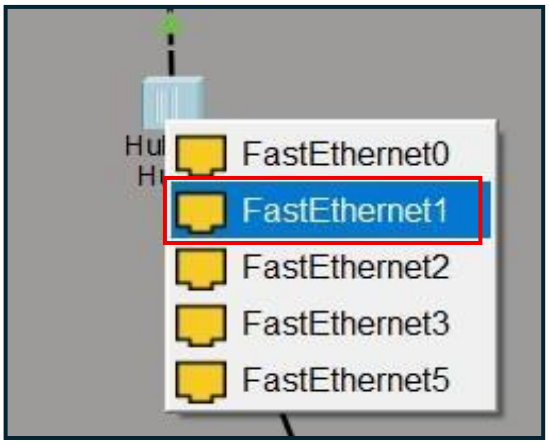


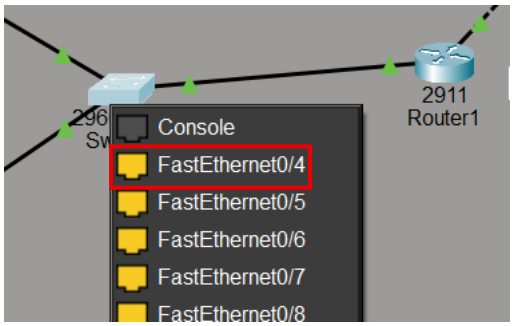
Fig-2: Pc/Laptop to Switch, and Switch to Router connection

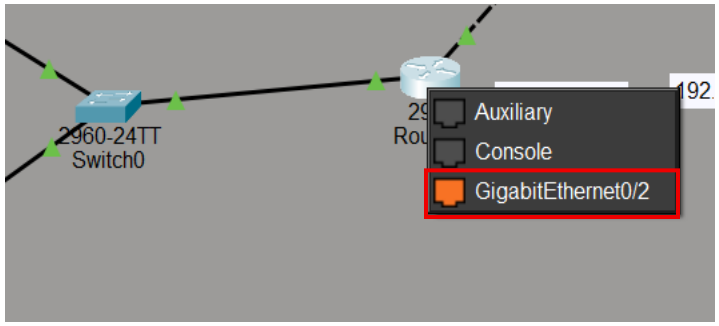
For Pc/Laptop to Switch Connection We use FastEthernet Port:-



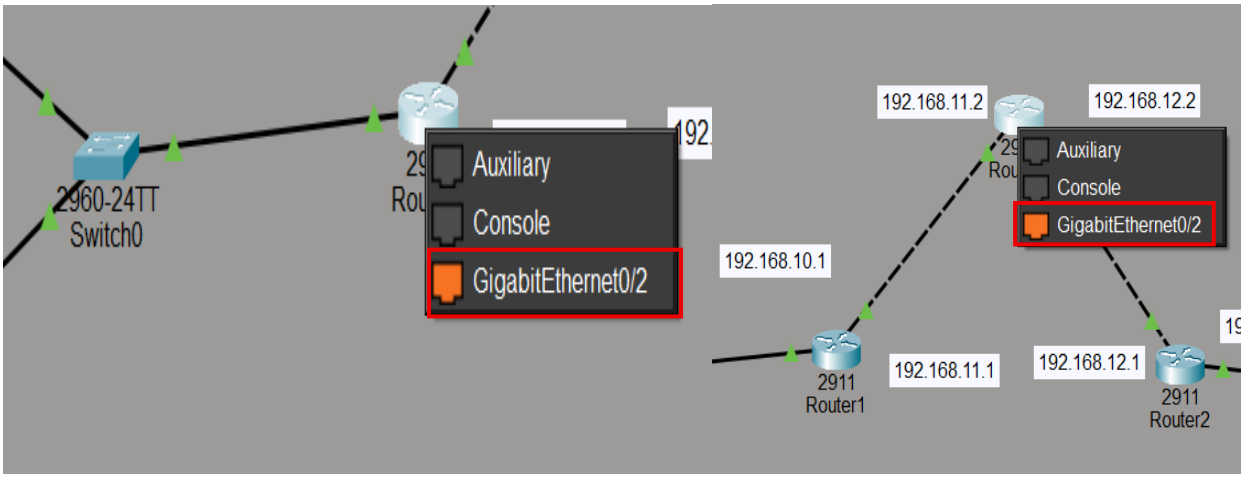


For Switch to Router Connection, we use FastEthernet Port to GigabyteEthernet Port: -

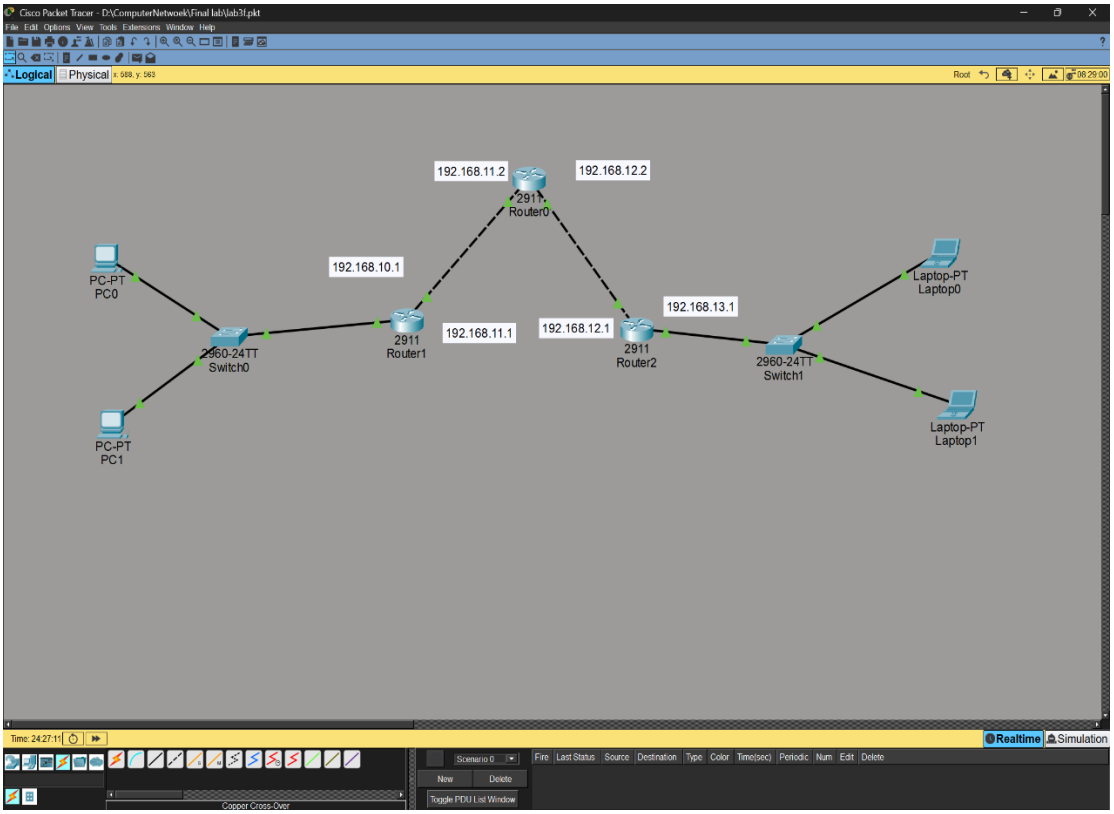




For Router-to-Router Connection, we use GigabyteEthernet Port to GigabyteEthernet Port: -

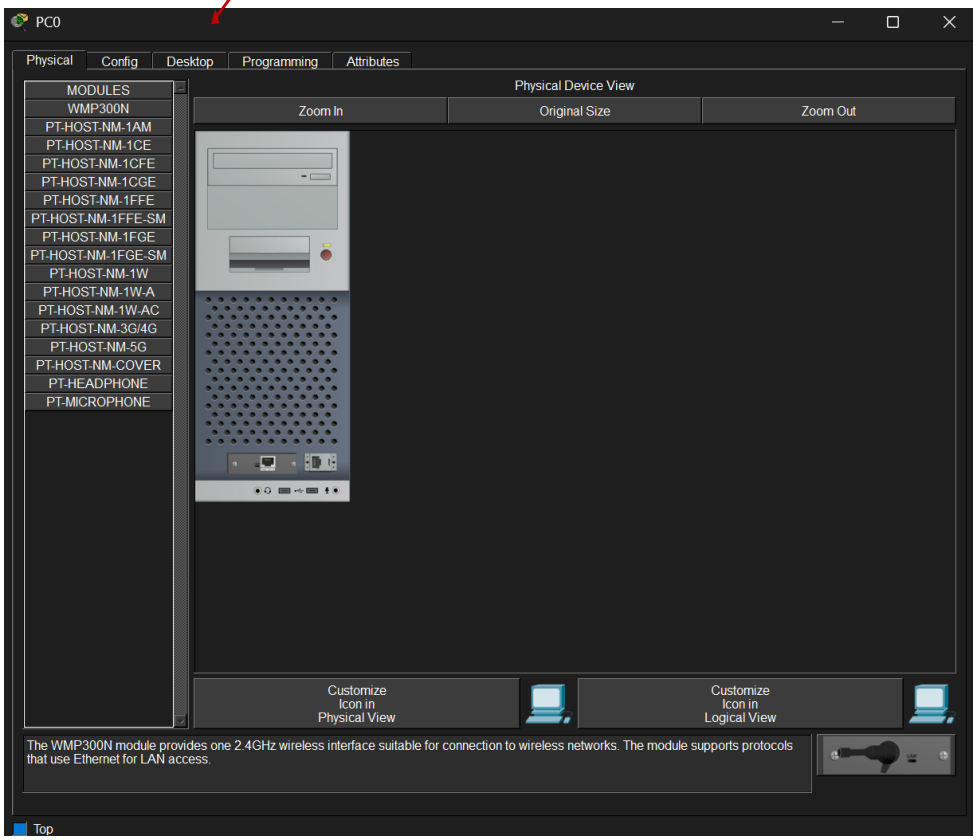


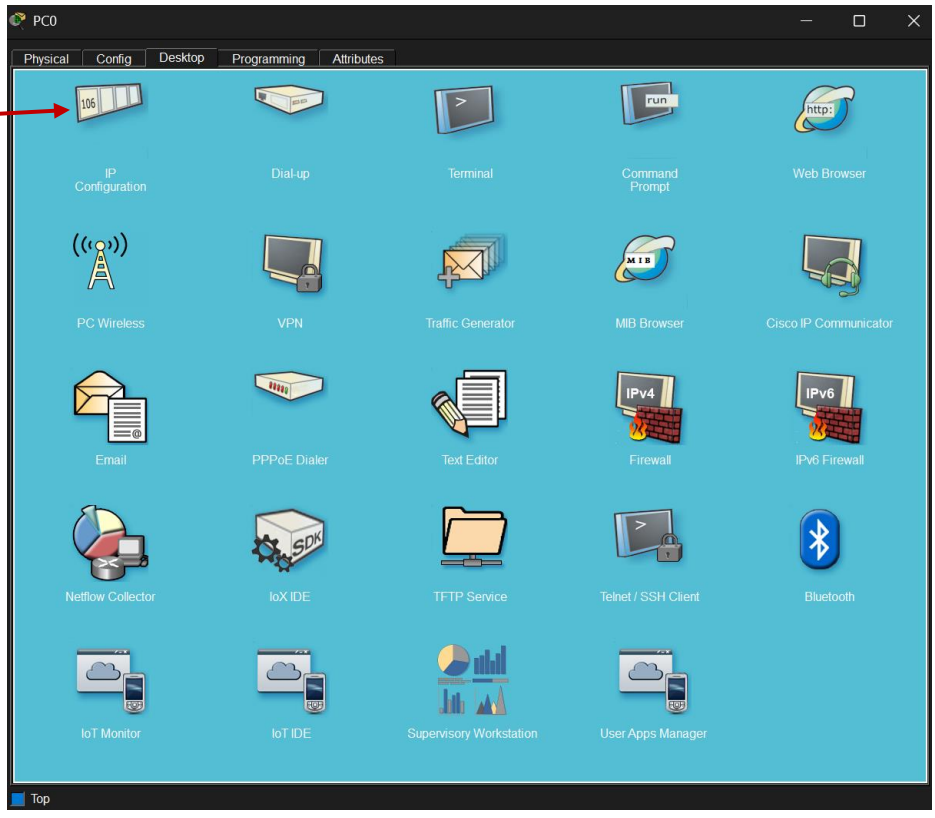
After Connection the Component My Final Design is:

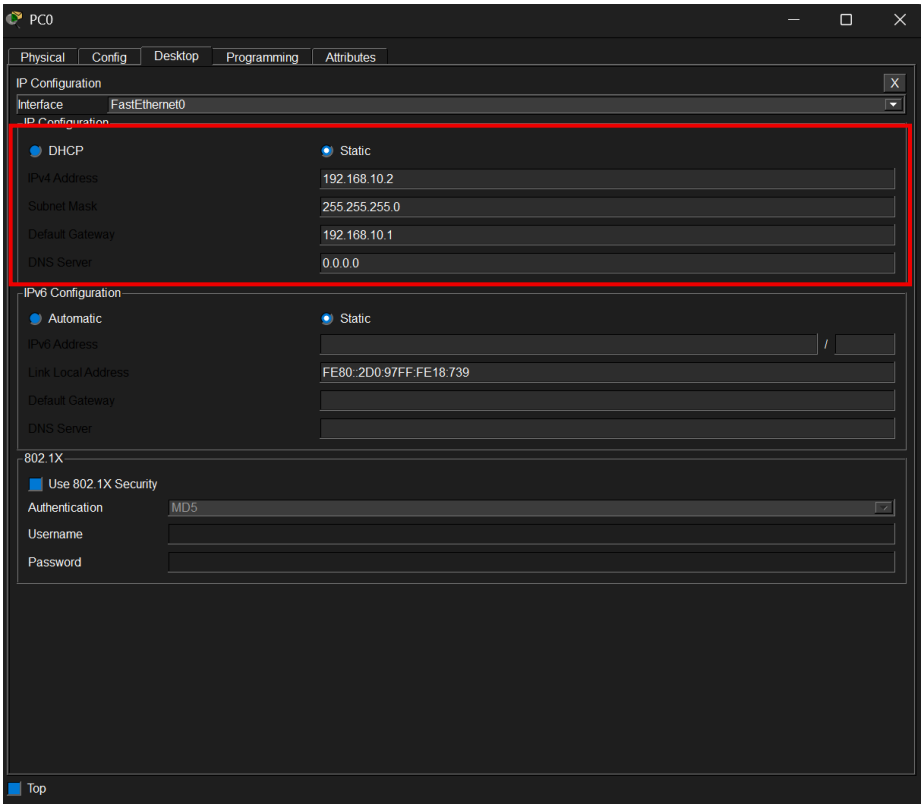


4. Step:4: - Pc/Laptop Ip configuration:

Click pc→ Desktop→ IP Configuration→Set Ip







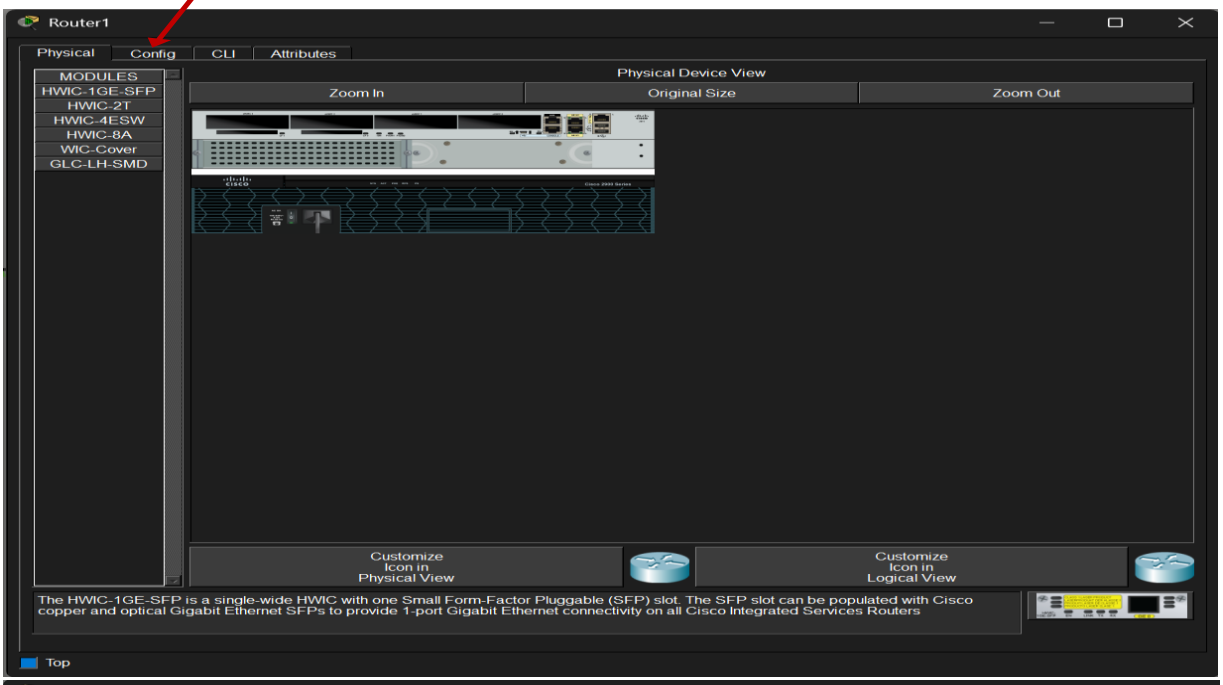
Select IP Address Static IPv4 Address: - 192.168.10.2 Subnet Mask :255.255.255.0 Gateway: - 192.168.10.1

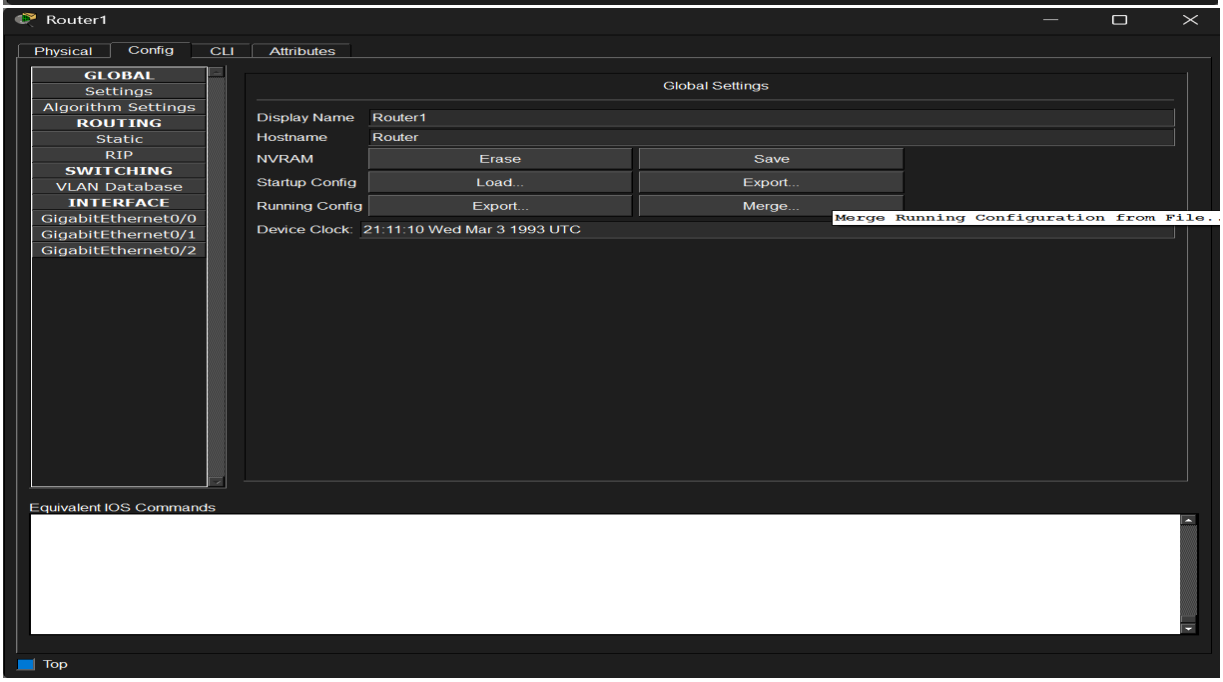
Then Same process for all Pc and Laptop for set Ip address and default Gateway.

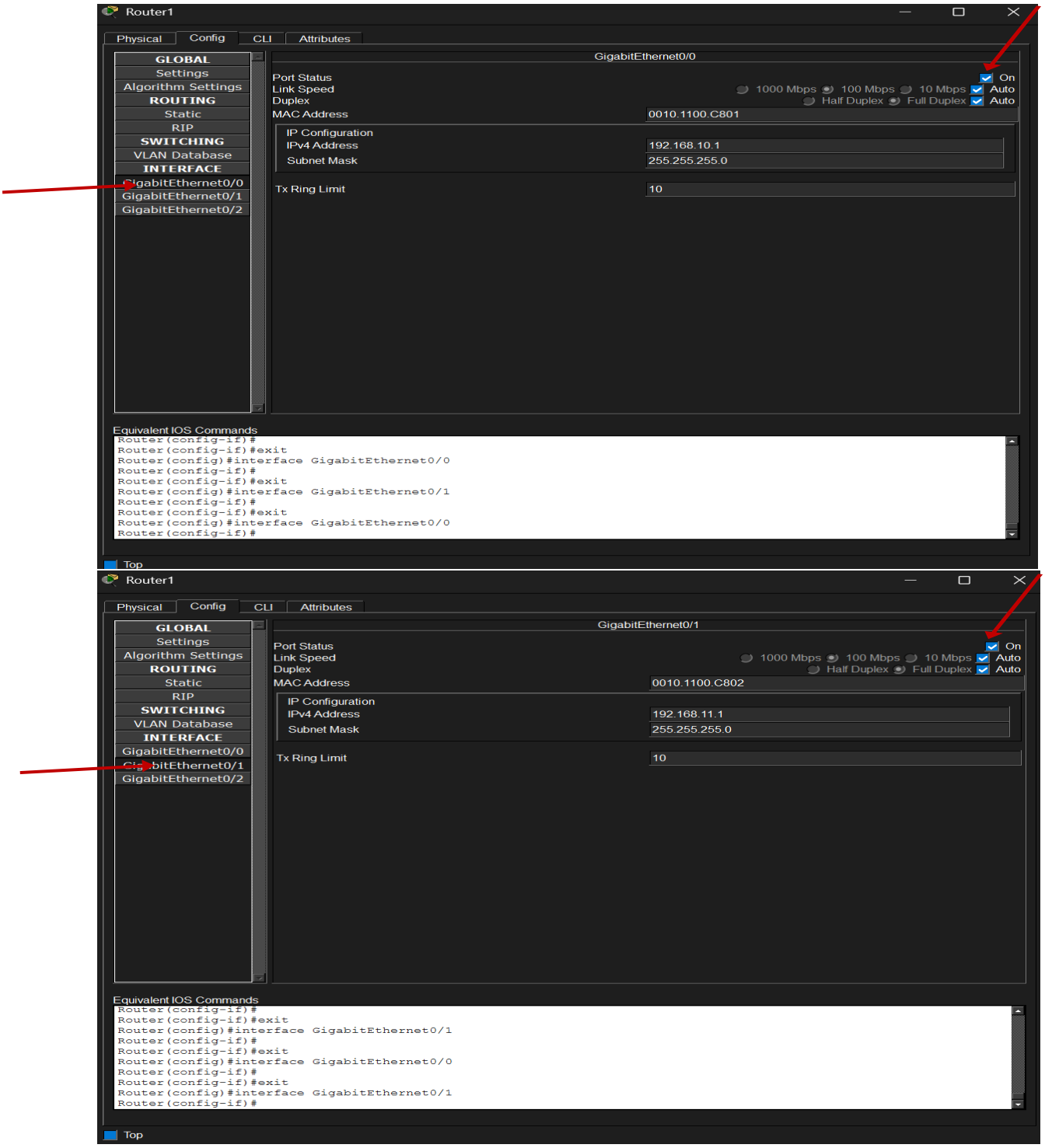
Device Name	IP Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.3	255.255.255.0	192.168.10.1
Laptop0	192.168.13.2	255.255.255.0	192.168.13.1
Laptop1	192.168.13.3	255.255.255.0	192.168.13.1

5. Step-5: - Router configuration:

Click Route→Config→
GigabyteEthernet0/0→
GigabyteEthernet0/1→On







Here for this Router GigabyteEthernet0/0 IP Address: - 192.168.10.1 Subnet Mask :255.255.255.0 and,

GigabyteEthernet0/1 IP Address: - 192.168.11.1 Subnet Mask :255.255.255.0

Then do same process for other two router

Note: Router On is must mandatory

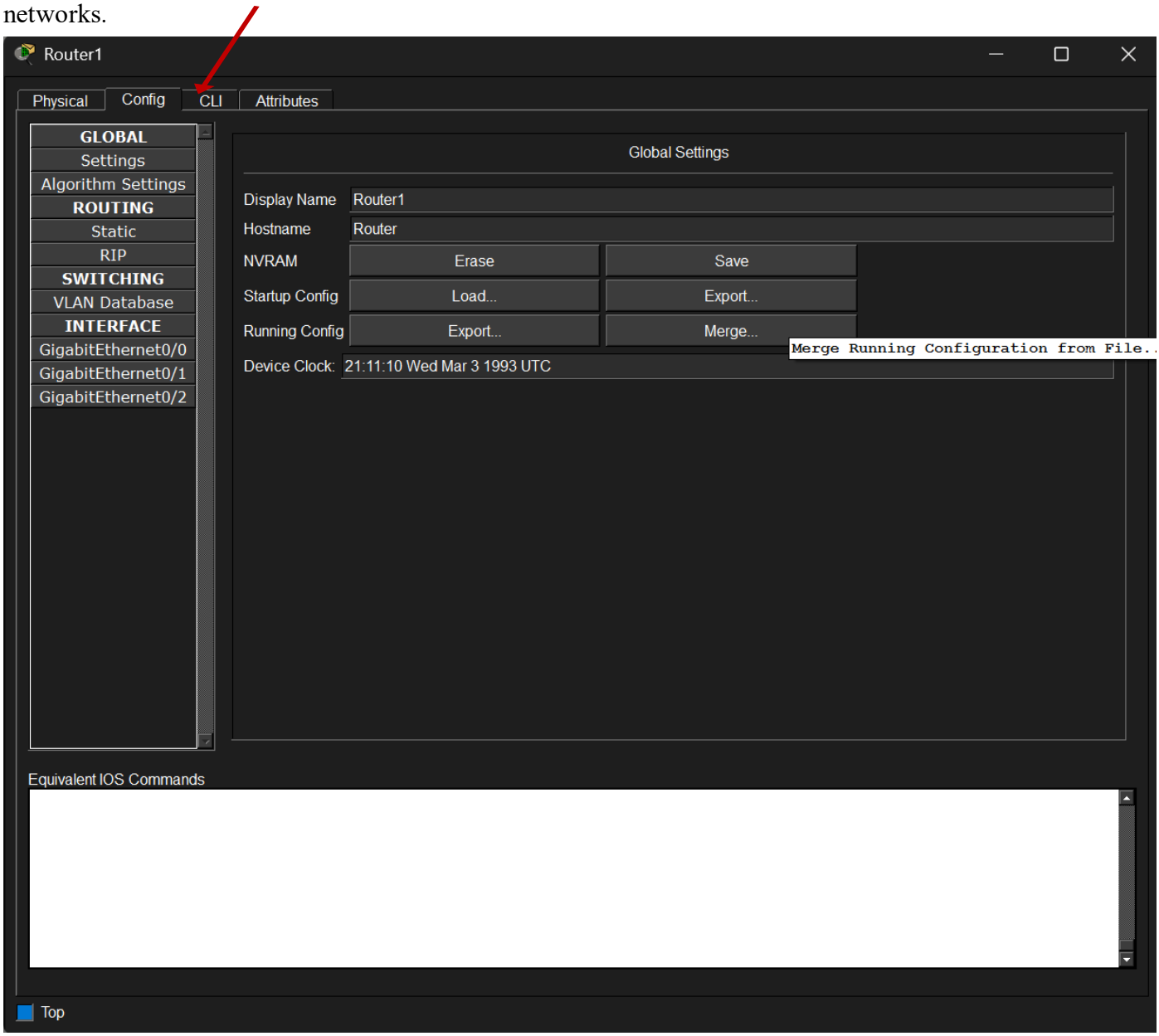
Device Name	GigabyteEthernet0/0	GigabyteEthernet0/1	Subnet Mask
Router0	192.168.11.2	192.168.12.2	255.255.255.0
Router1	192.168.10.1	192.168.11.1	255.255.255.0
Router2	192.168.12.1	192.168.13.1	255.255.255.0

6. Step-6: - Router CLI configuration:

After assigning the IP addresses to all router interfaces, configure RIPv2 from the CLI tab so that every router can learn the remote networks automatically.

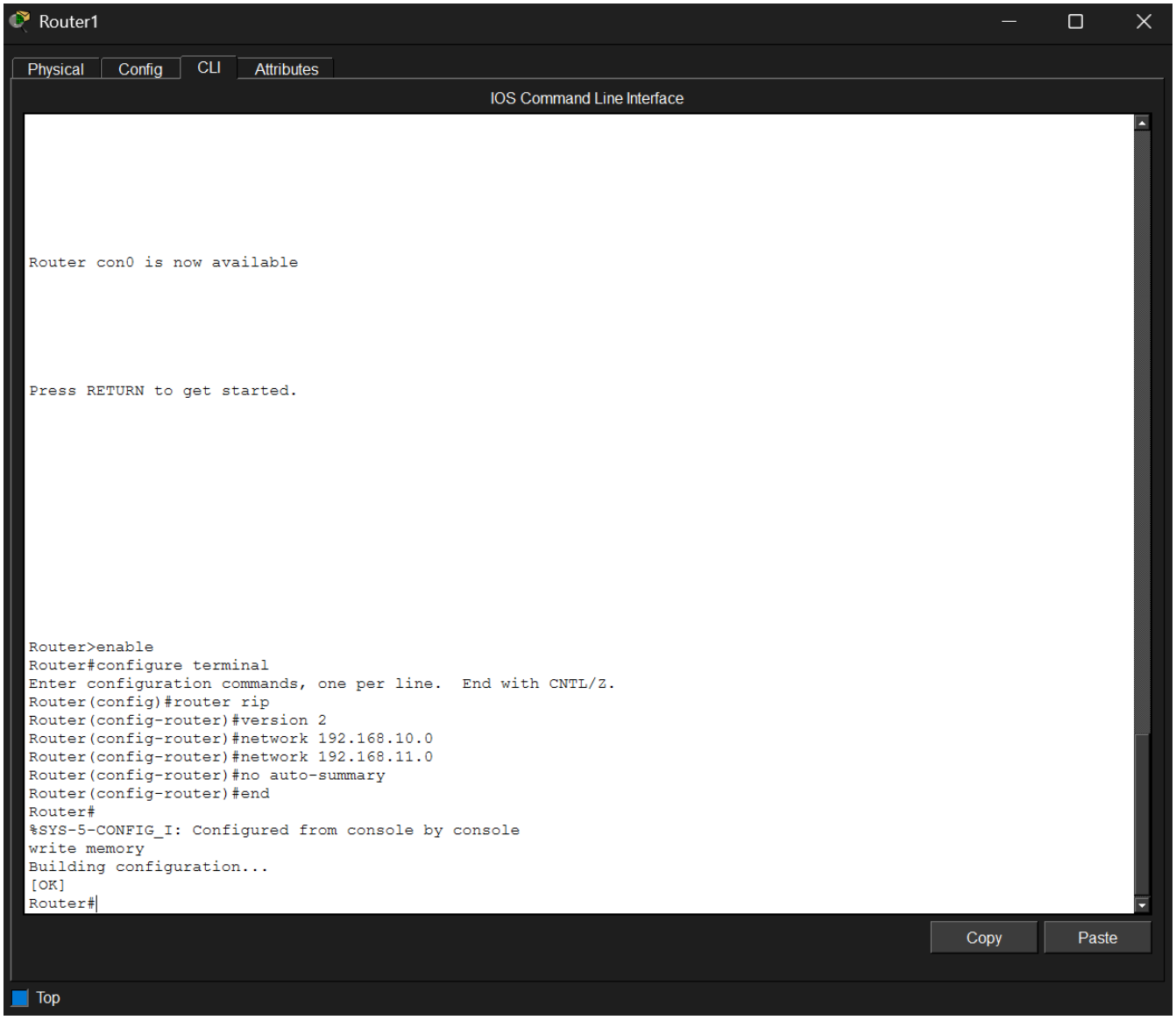
Open CLI: Click a router, open the CLI tab, and press Enter to access the command prompt.

Enter commands: Use the following RIPv2 commands on each router according to its directly connected networks.



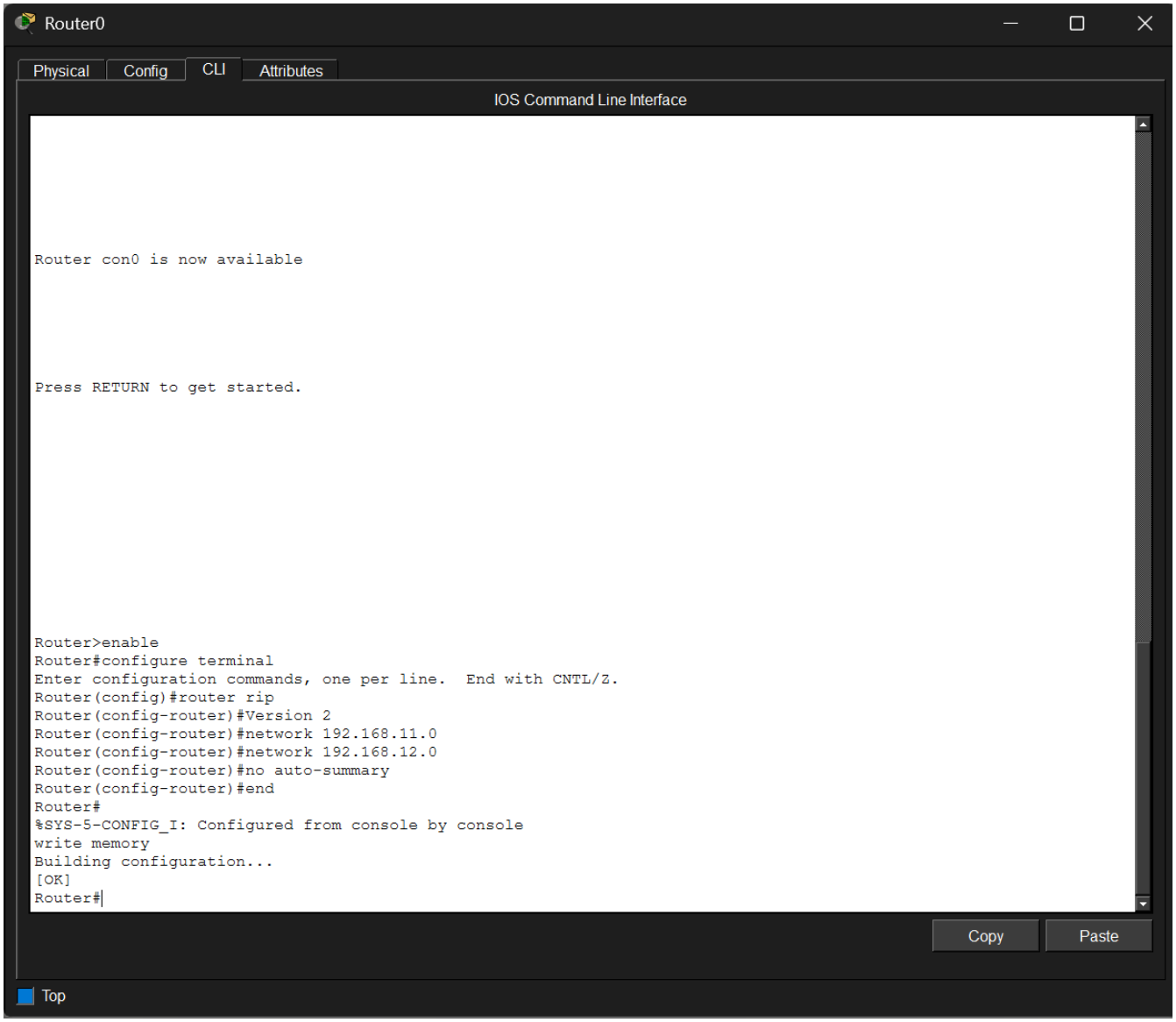
Router1 CLI Configuration

Router1 advertises the 192.168.10.0 LAN and the 192.168.11.0 router-to-router network.



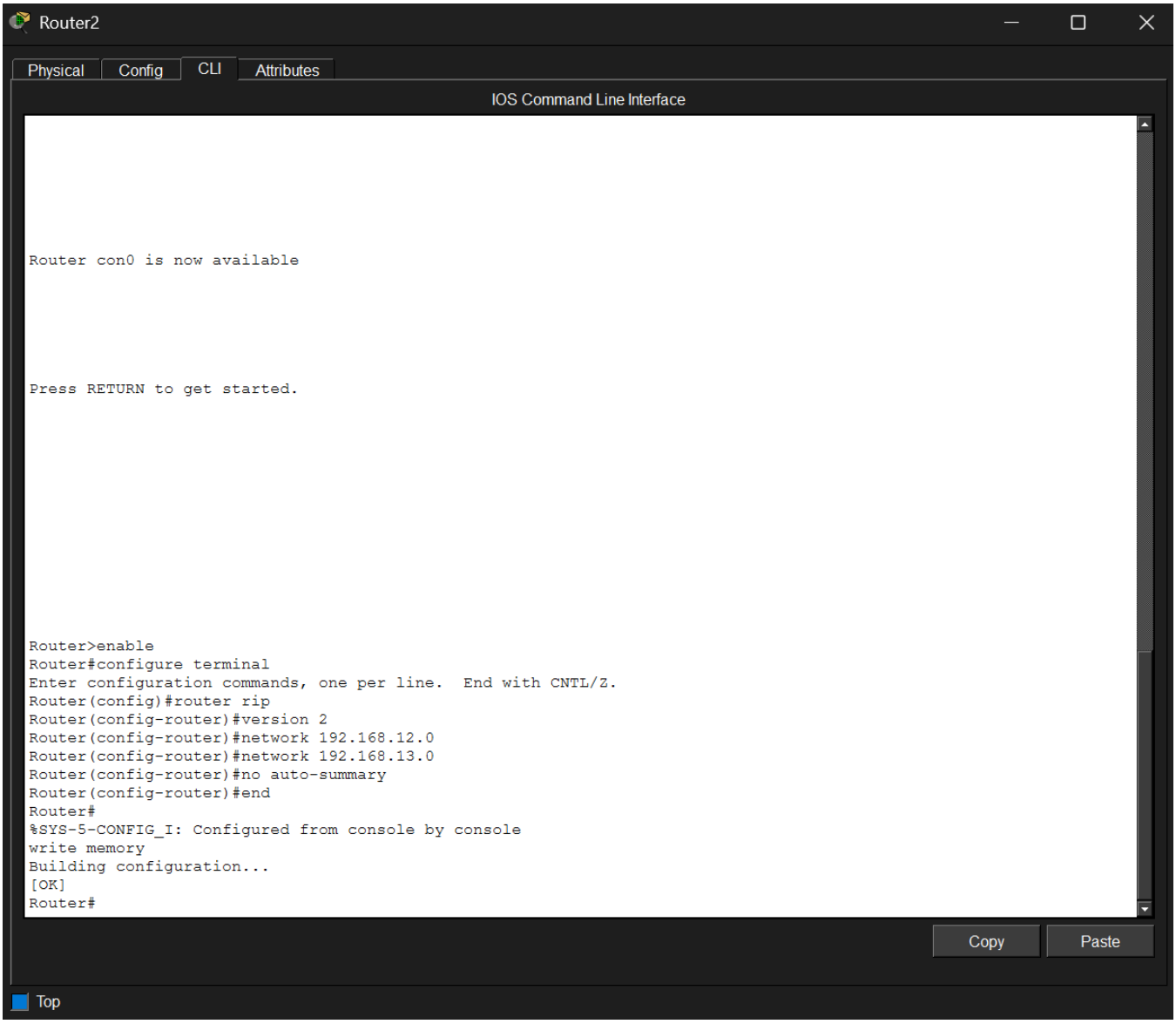
Router0 CLI Configuration

Router0 advertises the 192.168.11.0 and 192.168.12.0 router-to-router networks.



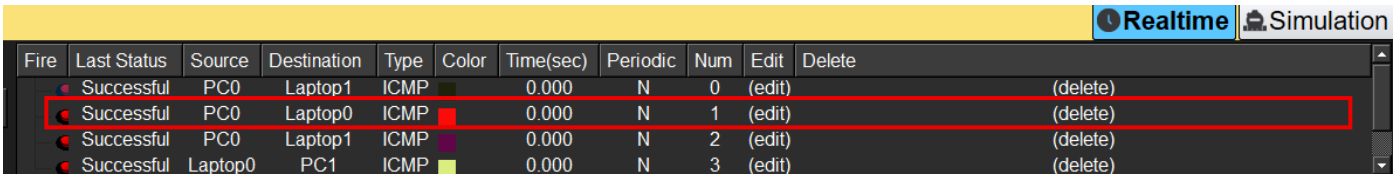
Router2 CLI Configuration

Router2 advertises the 192.168.12.0 router link and the 192.168.13.0 LAN.

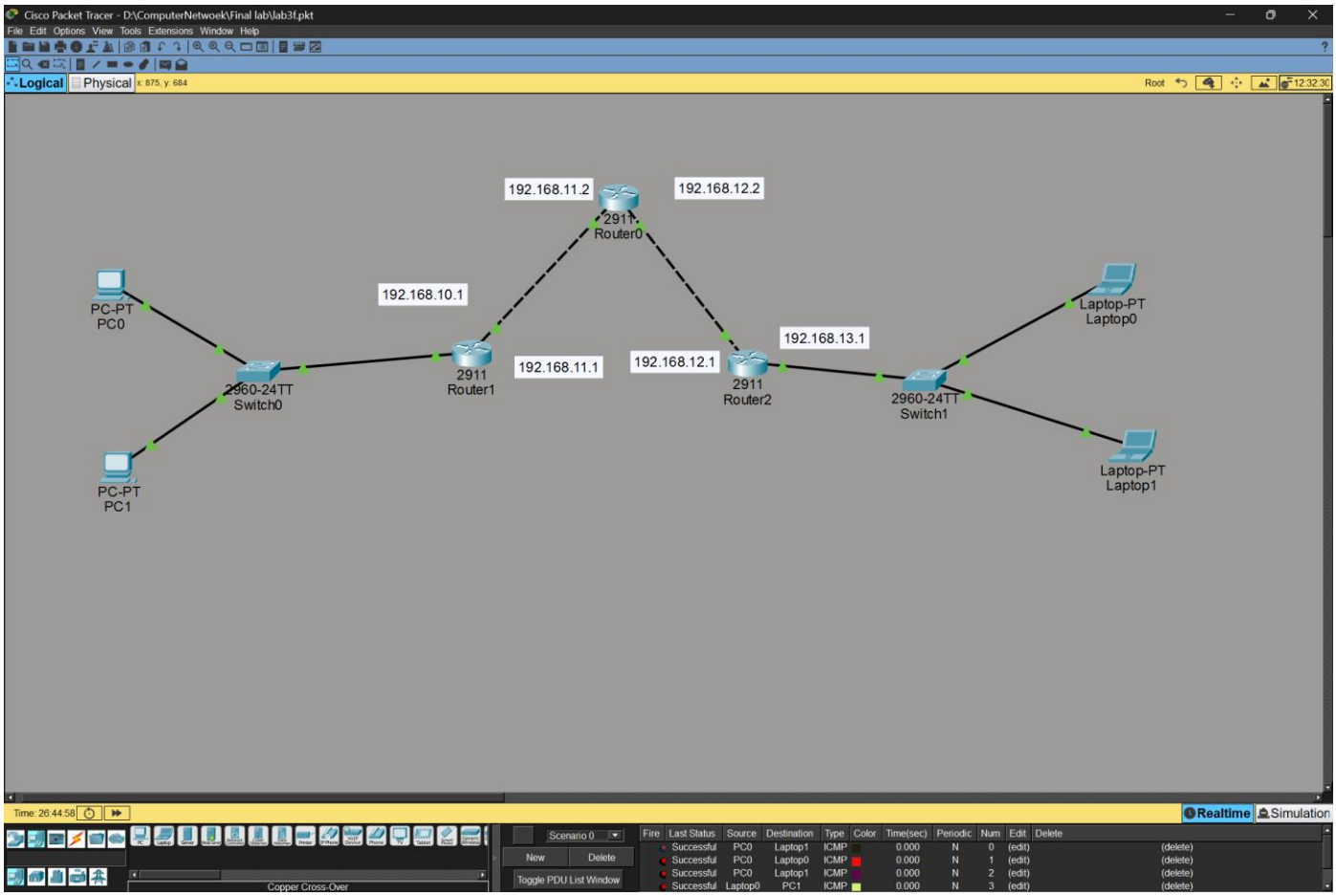


Result:

Finally, test end-to-end communication by sending a Simple PDU or by pinging 192.168.13.2 from PC0 and 192.168.10.2 from Laptop0.



Final view:


Cisco Packet Tracer - D:\ComputerNetwoek\Final lab\lab3f.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
Root
192.168.11.2
2911
Router0
192.168.12.2
PC-PT
PC0
192.168.10.1
Laptop-PT
Laptop0
192.168.13.1
2911
Router1
192.168.11.1
192.168.12.1
2911
Router2
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC1
Laptop-PT
Laptop1
Realtime
Simulation
Copper Cross-Over
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Successful PC0 Laptop1 ICMP 0.000 N 0 (edit) (delete)
Successful PC0 Laptop0 ICMP 0.000 N 1 (edit) (delete)
Successful PC0 Laptop1 ICMP 0.000 N 2 (edit) (delete)
Successful Laptop0 PC1 ICMP 0.000 N 3 (edit) (delete)